UAS SafeNet PHEFT2 HSM Deployment Guide - 6.0.1-GA (AI)

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1. Preface - UAS SafeNet PHEFT2 HSM Deployment Guide

This guide contains instructions on the installation and setup of SafeNet PHEFT (ProtectHost Electronic Fund Transfer) 2.0.0 Hardware Security Module (HSM).

Intended Reader

This document is intended as reference for administrators.

Document Scope

This guide covers the first-time setup and configuration of the SafeNet PHEFT2 HSM.

For any feedback or questions regarding this document, contact doc@i-sprint.com.

Related Documents

For more information, see the following documents in the AccessMatrix UAS (Universal Authentication Server) documentation set:

- *AccessMatrix Release Notes*
- *AccessMatrix Common Administration Guide*
- *UAS Administration Guide*

2. Introduction to UAS SafeNet PHEFT2 HSM Deployment

PHEFT1 and PHEFT2 Mixed Deployment

PHEFT2 is using newer firmware compared to PHEFT1. Although Safenet supports an environment that use both PHEFT1 and PHEFT2, please take note of the following:

PHEFT1 firmware	SHA-2 Usage	Supported	Action
Supports SHA-1 only (older than M090400)	N/A	Yes	None
Supports SHA-2 (since firmware version M090400 and older than M090800)	Yes	No	• Upgrade to latest firmware • Upgrade AM5 to version 5.1.2-SP1 and above
	No	Yes	None
Supports SHA-2 (firmware M090800 or newer)	Any	Yes	None

Notes:

- All the data above are true if there are no changes in AM server and AM E2EE JNI library
- All the data above are true assuming there's no change of feature used, e.g. character exclusion, SHA-2.

What's New in PHEFT2

Feature	Description
Hardware	
New Hardware	The PHEFT2 appliance has a new cryptographic module, dual power supply and an emergency decommission button.
Operational	
New Management Interfaces	PHEFT2 has two new console interfaces: LunaSH and Luna EFT Administration Console.
New Performance Level	Performance level has been increased to 2000 PIN translations per second.

Feature	Description
Security	
Separation of Duties	PHEFT2 enforces strong separation of duties to control access to sensitive services and functions.
Two-Factor Authentication	PHEFT2 provides two-factor authentication with dual user control. SafeNet eTokens have been introduced for authentication.
Secured Data Monitoring	PHEFT2 continues to provide secured data monitoring using secure audit logging, self-test and SNMP. Audit logging and self-test modules have been enhanced.

New Hardware

The release of PHEFT2 includes new hardware, such as:

- New cryptographic module.
- Redundant, hot-swappable power supplies.
- Emergency Decommission button – Pressing this button securely erases all the material stored on the unit.

New Management Interfaces

The use of the command-line and web console interface simplifies the process of configuring and managing PHEFT2:

- LunaSH is a command-line interface (CLI) that is used to perform administrative functions either at the serial console or remotely using SSH.
- Luna EFT Administration Console is a TLS secured web user interface that is used to perform management operations.

> **Note**: For further information, please refer to Management Interfaces section.

New Performance Levels

- PHEFT2 is now available at a maximum Performance Level (PL) of 2000 PIN translations per second. The different PLs supported by PHEFT2 are: PL 60, 140, 280, 1200, 2000. PL is in-field upgradable. Please check with SafeNet's Technical support team if you wish to upgrade the PL.

Once approved, the support team will provide you secure update package file with desired PL that can be installed on PHEFT2.

Separation of Duties

PHEFT2 enables multiple users to control access to sensitive services and functions. Please refer to Multiple Users and Roles section.

Two-Factor Authentication

PHEFT2 enforces two-factor authentication with dual user control to provide stronger security. The physical keys have been replaced by SafeNet eTokens that are protected with a user-configurable PIN.

Secured Data Monitoring

- Audit Logging: The new Audit Logging module records all security sensitive operation performed on HSM. A separate audit user has been added in PHEFT2 to manage secure audit logging. Audit log event records are stored in a file, with cryptographic safeguards ensuring verifiability, continuity, and reliability of log files.
- Simple Network Management Protocol (SNMP): PHEFT2 provides support for SNMP version 3 to retrieve vital information and receive notification or traps.

Management Interfaces

PHEFT2 provides two important interfaces to do management and configuration: **LunaSH** and **Luna EFT Administration Console**. Below table summaries components needed for both management interfaces.

User Interfaces	Required Component	Version/Description
LunaSH	Putty	Executable file for ssh/serial access
	Pscp	Secure Copy Program (SCP) for Windows command line processor
Luna EFT Administration Console	Operating System	Windows 7, 8
	Web Browsers	• Google Chrome v35 and above • Internet Explorer v11 and above • Mozilla Firefox v29 and above

User Interfaces	Required Component	Version/Description
	Safenet Authentication Client (SAC)	v9
	JAVA	1.8

> **Note**: You are not advised to log in to Luna EFT Administration Console and LunaSH at the same time. This prevents synchronization issues that could arise from multiple users changing the configuration simultaneously.

eToken

eToken devices are distributed together with PHEFT2 HSM. They act as PIN to separate duties among users. eToken devices are manage by a middleware -SafeNet Authentication Client (SAC). For more information about SAC, please refer to the G. SafeNet Authentication Client (SAC) section.

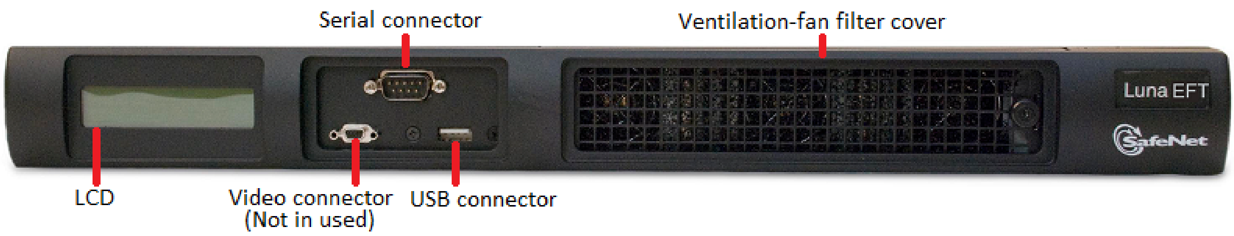
eToken	Quantity	eToken Name	Duty
Purple	1	Activation eToken	Performs HSM Activation
Blue	2	EFT Administrator(s) eToken	Performs administrative tasks
Black	2	Partition Owner(s) eToken	Performs cryptographic tasks
White	2	EFT Auditor(s) eToken	Performs audit operations

Multiple Users and Roles

Users & Roles		User Interface	Authentication	Description
OS Users	admin (also known as appliance admin)	LunaSH	Password-based	An admin is required to perform general appliance-level administration such as network and date/time configuration.
	audit	LunaSH	Password-based	An auditor is required to secure a copy of authorized archived logs from PHEFT2.

Users & Roles		User Interface	Authentication	Description
Roles	EFT Administrator(s) or EFT Admin(s)	LunaSH / Luna EFT Administration Console	eToken (Blue)	Two EFT admins are required to perform administrative tasks such as partition creation, network settings, printer configuration, and certificate management for Luna EFT Administration Console.
	Partition Owner(s)	LunaSH / Luna EFT Administration Console	eToken (Black)	Two EFT partition owners are required to perform cryptographic operations such as key management, PIN security settings, key settings and certificate management for secure channel.
	EFT Auditor(s)	LunaSH	eToken (White)	Two EFT auditors are required to perform audit operations such as audit-related key management and archival of audit logs.

Front Panel



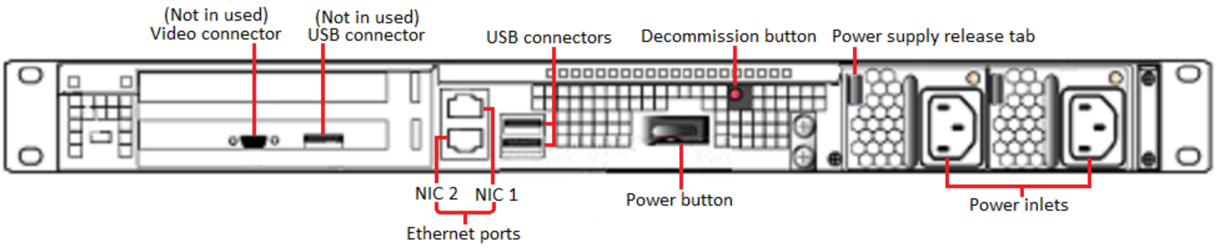
The front interface has five components:

- **LCD** - Displays information on the current operational status of the appliance during different operations.
- **Video connector** (Not in use) - The 15-PIN VGA connector enabled via the console switch provides a VGA video output for an analog VGA monitor.
- **Serial connector** - The serial connector is reserved for administration computer connections. It is recommended that you have a null-modem serial connection between the serial console port

and the administration computer or a terminal. This is for convenience during the initial setup, so your administrative connection remains active when you assign new IP addresses.

- **USB connector** - The USB connector present on the front panel is designated for keyboards, eTokens for authentication, smartcard readers and printers for PIN printing via a supplied USB-to-Serial converter.
- **Ventilation-fan filter cover** - The removable bracket allows the cleaning of the air filter.

Rear Panel



The rear Interface has seven components:

- **Video connector** (Not in use) **-** The 15-PIN VGA Connector enabled via the console switch provides a VGA video output for an analog VGA monitor.
- **USB connectors** - Can optionally be used with the same peripherals as the front USB socket.
- **Ethernet ports** - For network connection of your PHEFT2 appliance.
- **Power button** - Use to stop the system if the command-line shutdown is not available; use to restart the system if it has been switched off.
- **Decommissioning button** - Recessed for safety; renders PHEFT2 contents unusable.
- **Power supply release tab** - Press tab to release the catch, and slide the power supply out.
- **Power inlets** - Two power inlets are provided. For proper redundancy and best reliability, the power cables should connect to two completely independent power sources.

3. Configuring SafeNet PHEFT2 HSM

	Procedure	Required User/ Role
HSM basic configuration		
1.	Power On PHEFT2 Appliance and connect it to a dumb terminal or a PC through serial port. Please refer to the Establishing a Connection section.	admin
2.	Set date and time. Please refer to the Setting Date and Time section.	admin
3.	Initialization prepares the HSM for use by setting up the necessary identities and ownership (EFT Activation, EFT Admins, EFT Auditors) that are to be associated with the HSM. Please refer to the Initializing PHEFT2 HSM section.	admin, EFT Admins, EFT Auditors
4.	Activate PHEFT2 by verifying the activation eToken. Please refer to the Activating PHEFT2 HSM section.	admin, EFT Admins
5.	Configure network setting and access control. Please refer to the Configuring Network Interface section.	admin, EFT Admins
6.	Generate certificate to use Luna EFT Administration Console. Please refer to the Generating Luna EFT Administration Console Certificate section.	admin, EFT Admins
7.	Configure audit logging to securely log details of transaction, data, events and sensitive operations. Please refer to the Configuring Audit Logging section.	audit, EFT Auditors
8.	EFT Admin to enabling or disabling of features that are to apply to the PHEFT2. Please refer to the Setting PHEFT2 HSM Policies section.	admin, EFT Admins
9.	Create HSM partition to separate cryptographic work-spaces that reside within the PHEFT2 appliance. Please refer to the Creating PHEFT2 HSM Partition section.	admin, EFT Admins

	Procedure	Required User/ Role
10.	Set up host communications. Please refer to the Setting HSM Client White List section.	admin, EFT Admins, EFT Partition Owners
11.	Configure Security Module Host Comm in the AccessMatrix Server; please refer to the Configuring Security Module Host Comms section.	AM system Admin
Additional steps for HSM used for password generation		
12.	Store smartcard Transfer Protect Key. If you would like to restore keys from smartcard in the next step, please use the same Transfer Protect Key used to backup to the smartcard. Please refer to the SmartCard Transfer Protect Key (KTP) section.	admin, EFT Partition Owners, key Custodians
13.	Restore keys from smartcard, if necessary. Please refer to the Retrieving or Restoring Keys from SmartCard section. If you had restored keys from smartcard, you may want to skip step 13 and step 15.	admin, EFT Partition Owners, Smartcard Owner
14.	Store Transformed PIN Value Key (KTPV). Please refer to the Creating Transformed PIN Value Key (KTPV) section.	admin, EFT Partition Owners, key Custodians
15.	Store PIN Protect Key (PPK). Please refer to the Creating PIN Protect Key (PPK) section.	admin, EFT Partition Owners, key Custodians
16.	Store Domain Master Key (KM), if required for AM5 Key Managment. Please refer to the Domain Master Keys (KM) section.	admin, EFT Partition Owners, key Custodians
17.	Store Data Protect Key (DPK), if required for AM5 Key Managment. Please refer to the Creating Data Protect Key (DPK) section.	admin, EFT Partition Owners, key Custodians
18.	Configure RSA. Please refer to the Managing HSM RSA Keys section.	admin, EFT Partition Owners
19.	Enable the transferable HSM RSA keys configuration. Please refer to the Enabling Transferable HSM RSA keys section.	EFT Partition Owners
Additional steps for HSM used for password printing		

	Procedure	Required User/ Role
20.	Store PPK if you have not done so in step 18. Please refer to the Creating PIN Protect Key (PPK) section.	admin, EFT Partition Owners, key Custodians
21.	Configure printer port. Please refer to the Configuring Printer Setting section.	EFT Admins
22.	Align printer. Please refer to the Printing Alignment (Printing Parameters) section.	EFT Partition Owners
23.	Set password restriction. Please refer to the Configuring Password Restriction section.	EFT Partition Owners
24.	Enable or disable OBM Password Mailer, if necessary. Please refer to the Configuring OBM Password Mailer Setting section.	EFT Partition Owners
Additional steps for other purposes		
25.	If failover load balancing is required, then please refer to the Failover Load Balancing section.	EFT Partition Owners
26.	Poweroff or reboot PHEFT2 HSM, then please refer to the Poweroff or Rebooting PHEFT2 HSM section.	admin

3.1. Configuring PHEFT2 HSM

This section describes how to configure the PHEFT2 HSM, such as configuring network access and changing the access control password.

Establishing a Connection

To open a connection:

1. Connect a null-modem serial cable (supplied) between the serial port on the PHEFT2 appliance front panel and a dumb terminal or a PC (for example a laptop) that will serve as the administration computer. Once network parameters are established, you can switch to an SSH session over your network.
2. When a successful connection is made, a terminal window opens and the prompt "login as:" appears.
3. Login as admin. Please refer to the C.1 Admin User Login LunaSH section.
4. You are prompted for the admin password. If this is the first time you have connected, the default password is "PASSWORD", and you are required to change it to something more secure.

Once you have logged in, the system presents the Luna Shell prompt: `[local_host] lunash:>`

Setting Date and Time

For setting date and time of the PHEFT2 HSM, please do the following steps:

1. From LunaSH command, login as admin. Please refer to C.1 Admin User Login LunaSH section.
2. Verify the current date, time and timezone on PHEFT2. At the lunash prompt, type the command:
   `lunash:> status date`
   `lunash:> status time` or `lunash:> status timezone`
3. If the date, time or timezone are incorrect for your location, change them using `sysconfig` command:
   For example:
   `lunash:>sysconfig time 14:30 20150606`
   Time set to Fri Jun 6 14:30:00 EDT 2015
   `lunash:> sysconfing timezone set Asia/Singapore`
   Timezone set to Asia/Singapore

> **Note**: For more information on time zone abbreviations, please visit http://www.worldtimezone.com/.

Initializing PHEFT2 HSM

Initialization prepares the HSM for use by setting up the necessary identities and ownership that are to be associated with the HSM. You must initialize the HSM one time before you can generate or store keys or perform cryptographic operations.

To initialize PHEFT2, perform the following steps:

1. From LunaSH command, login in as admin. Please refer to C.1 Admin User Login LunaSH section.
2. Type the following command:

   `lunash:>hsm eftinit`

   This command will perform the following tasks:

   - Initialize EFT Activation eToken
   - Initialize eToken for EFT Admins
   - Initialize eToken for EFT Auditors

> **Notes**:
> - You can re-initialize a Luna EFT at any time. Re-initialization destroys all data on the PHEFT2 and eToken. To view the impact of `hsm eftinit` command on PHEFT2, see F. Destructive Actions in Appendixes.
> - Refer to G. SafeNet Authentication Client (SAC) for eToken management.

Initializing EFT Activation eToken

The `hsm eftinit` command will prompt you to insert the activation eToken (Purple).

Perform the following steps:

1. Insert the Luna EFT activation eToken.
2. Enter the PIN; re-enter the PIN to confirm.

> **Note**: A PIN may consist of upper case and/or lower case letters, digits and printable special characters, but between eight and fifteen characters.

Initializing eToken for EFT Admins

The PHEFT requires dual administrator authentication for the console interface. It uses eToken (Blue) for authentication.

To initialize eToken for each EFT Admin:

1. Enter the EFT Admin name.
2. Insert the eToken for EFT Admin.
3. Enter the PIN; re-enter the PIN to confirm.

> **Note**: A PIN may consist of upper case and/or lower case letters, digits and printable special characters, but between eight and fifteen characters.

Initializing eToken for EFT Auditors

The PHEFT2 required dual auditor authentication for the console interface. It uses eToken (White) for authentication.

To initialize eToken for each EFT Auditor:

1. Enter the EFT Auditor name.
2. Insert the eToken for EFT Auditor.
3. Enter the PIN, and re-enter the PIN to confirm.

> **Note**: A PIN may consist of upper case and/or lower case letters, digits and printable special characters, but between eight and fifteen characters.

Activating PHEFT2 HSM

Before performing any key processing operations on PHEFT2, you have to first activate PHEFT2 by verifying the activation eToken.

Perform the following steps for eToken activation:

1. From LunaSH command, login as an admin. Please refer to the C.1 Admin User Login LunaSH section.
2. Type the following command:
   `lunash:>hsm activate`

3. Insert the activation eToken. The system prompts you to enter the PIN for activation eToken.
4. Enter the activation eToken PIN.
5. On successful activation, the message "HSM activated successfully" is displayed. The Luna EFT is now ready to perform cryptographic operations.

> **Note**: To enable PHEFT2 HSM auto activation for each reboot, please refer to Setting PHEFT2 HSM Policies section.

Configuring Network Settings

This section describes steps on network settings.

Configuring Network Interface

To configure network interface, perform the following steps:

1. From lunaSH command, login as an admin. Please refer to C.1 Admin User Login LunaSH section.
2. To display the current network interface settings, type the following command:
   `lunash:>sysconfig network interface show`
3. Then, log in as an EFT admin. Please refer to C.3 EFT Admin Login LunaSH section.
4. To set the hostname of the PHEFT2 appliance (use lowercase characters), type the following command:
   `lunash:>sysconfig network hostname`
5. Use network interface to change network configuration settings. For example:
   `lunash:>sysconfig network interface static -device eth0 -ip 172.25.15.176 -netmask 255.255.255.0 -gateway 172.25.15.3`
   a. If you choose to configure the second ethernet port (eth1), repeat the network interface command above, replacing the above values with `eth1` and the appropriate address for that device.
   b. If you choose to configure bond0 (multipathing), repeat the network interface command above while replacing the relevant values. For example:
      `lunash:>sysconfig network interface static -device bond0 -ip 172.25.15.176 -netmask 255.255.255.0 -gateway 172.25.15.3`
6. Use the `sysconfig network interface show` command to verify the current settings.
7. Verify correctness of your network setup by pinging another server using following command:
   `lunash:>sysconfig network interface ping <IP address>`

Generating Luna EFT Administration Console Certificate

Once you have configured the network settings, generate a SSL certificate/key pair for accessing Luna EFT Administration Console. PHEFT2 supports two modes of certification: CA-signed and Self-signed. The default mode is CA-signed certificate. To generating Luna EFT Administration Console certificate, perform the following steps:

1. From LunaSH command, login as an admin. Please refer to the C.1 Admin User Login LunaSH section.
2. Then, login as an EFT admin. Please refer to the C.3 EFT Admin Login LunaSH section.
3. To view the mode of certification, type the following command:
   `lunash:>sysconfig SSLMgmt viewmode`
4. To set the mode of certification, type the following command:
   `lunash:>sysconfig SSLMgmt setmode -mode <Self_Signed/CA_Signed>`
5. The command for generating certificate is:
   `lunash:>sysconfig certMgmt generate -modulus <string> -days <string> -subject <string>`
   - `modulus` - The modulus values are: 2048, 3072, 4096.
   - `days` - The number of days for which the certificate remains valid. The default validity period is 3650 days.
   - `subject` - The subject string include certificates attributes, i.e., `- CN <common name> - O <organization> - OU <unit> - C <country> -ST <state> -L <location> - EmailAddress <email>`
     If the custom subject is not specified, then the default subject string will be: "Subject:/CN=Luna EFT/O=Safenet Inc/OU=Information Technology/C=US/ST=Maryland/L=Belcamp/emailAddress=support@safenet-inc.com"

<u>Example of CA-signed Certificate Generation Process</u>

1. After performing steps 1 to 5 in the above section, export the generated CA cert into file in PHEFT2.
   `lunash:>sysconfig certMgmt export -type server`

i	**Note**: If the SSL mode has been set to self-signed, then self-signed certificate will be exported, otherwise, CSR will be exported.

2. To view the generated certificate, erent following command:

   ```
   lunash:>my file list
   ```

3. Run **pscp.exe** in client machine to get certificate from PHEFT2 for CA certificate signing, enter following command:

   ```
   pscp admin@<host_ip>:<cert_file_Name> <target_path>
   ```

4. Import CA cert and signed cert back into PHEFT2.

   ```
   pscp <cert_file_path> admin@<host_ip>:<target_path>
   ```

5. Import CA cert into browser.

6. To open Luna EFT Administration Console, please refer to C.6.1 EFT Admin Login Luna EFT Administration Console section.

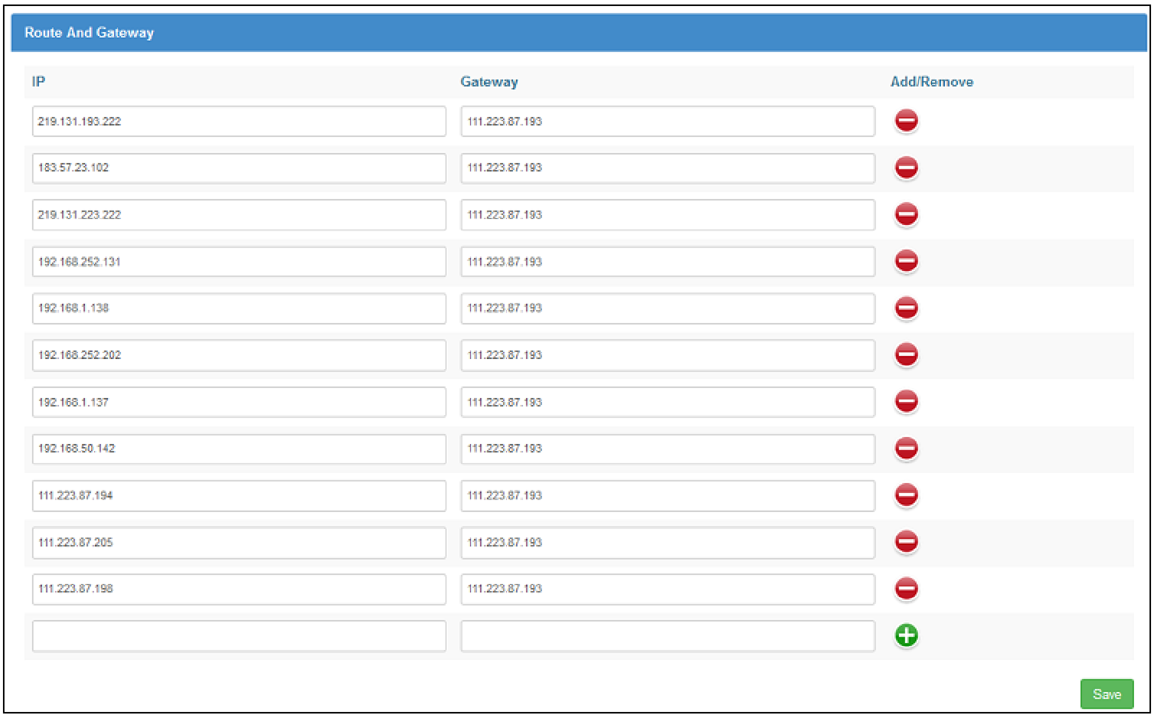
Setting HSM Client White List

Once you have configured basic network settings and generated a certificate for accessing Luna EFT Administration Console, login to the Luna EFT Administration Console. The Luna EFT Administration Console is a graphic user interface that enables you to perform remote administration using a web browser. For further details, please refer to the Console Guide.

To establish a host communication, perform the following steps:

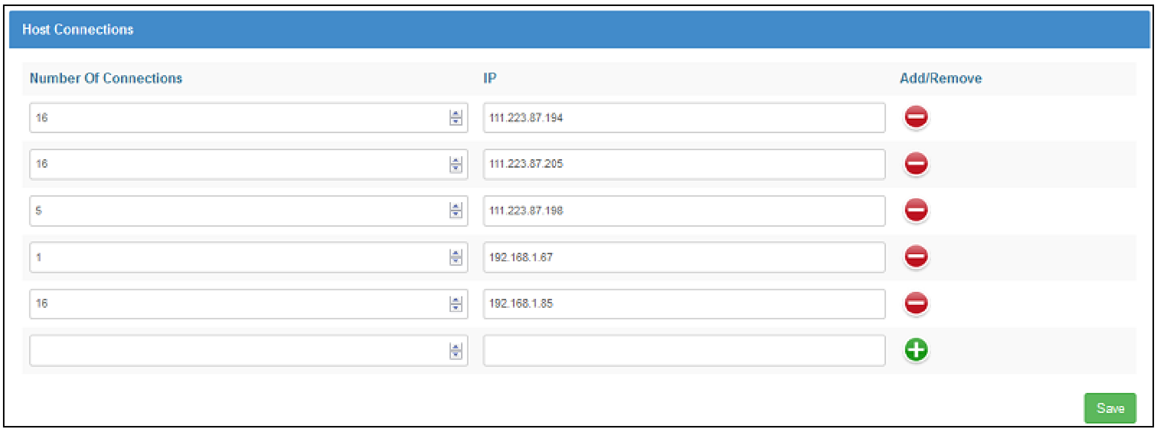
1. From Luna EFT Administration Console, login as an EFT Admin. Please refer to C.6.1 EFT Admin Login Luna EFT Administration Console section.

2. Go to **System Configuration > Network Configuration > Routing and Gateway** and configure the network settings so that PHEFT2 communicates to a host via a gateway.



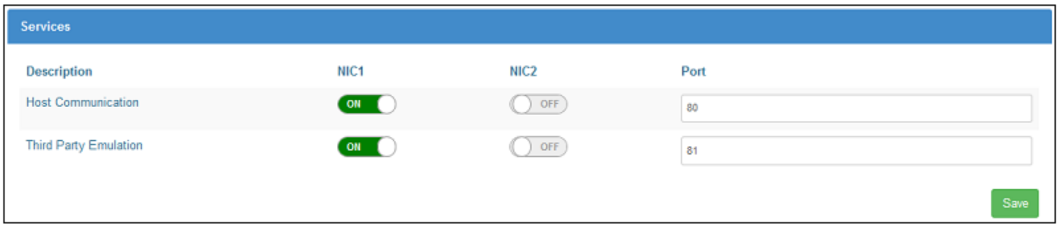
3. From Luna EFT Administration Console, login as EFT Partition Owner using HSM PartitionOwner eToken. Please refer to C.6.2 EFT Partition Owner Login Luna EFT Administration Console section.

4. Go to **System Configuration > Network Configuration > Host Connections** and register the host connections.



5. Then, go to **System Configuration > Network Configuration > Services** to:
   - Enable the relevant **NIC**(s) for **Host Communication** and enter the service port.

- Enable the relevant **NIC**(s) for **Third Party Emulation** and enter the service port.



Configuring Audit Logging

PHEFT2 provides cryptographic mechanisms to securely log details of transaction, data, events and sensitive operations so that these logs can be used for auditing. Audit Logging is designed to support administrators/auditors to figure out faults that may have occurred in the software. The audit logging is managed by an audit user, in combination with the EFT Auditor role, through a set of `lunash` commands. The EFT Audit role does not exist until it is created (initialized) during HSM initialization.

The appliance audit user is a standard user account on PHEFT2, with default password "PASSWORD". Audit role allows complete separation of audit responsibilities from the Appliance Admin, EFT Admin and the Partition Owner. If the Audit role is initialized, the EFT Admin(s) and Partition Owner(s) are prevented from working with the log files, and auditors are unable to perform administrative tasks on PHEFT2.

Assuming you have already initialized the Audit user role, perform the steps below:

1. From LunaSH command, login as an audit user. Please refer to the C.2 Audit User Login LunaSH section.
2. The first time you log into the Luna appliance as the user "audit", you are prompted to change the password, from default "PASSWORD" to something more secure.
3. Then, login as an EFT Auditor. Please refer to C.5 EFT Auditor Login LunaSH section.
4. Use the `?` option to see the top-level commands.
   `lunash:>?`
5. Generate the MAC key for audit logging. Type in the command:
   `lunash:>keyMgmt generate hsm mac`
6. Activate the MAC key. Type in the command:
   `keyMgmt activate mac`

i	**Notes**:

- Auditor can store 10 TDES double or triple length keys inside PHEFT2. These keys are used to generate MAC over audit logs generated by PHEFT2. Auditor can use these keys to verify integrity of audit logs exported out of PHEFT2.
- Only one index can be set as active at a time. Active key is used to generate MAC on audit logs. Although only active MAC key is used for MAC generation, other keys can be used for verification of older logs.

Setting PHEFT2 HSM Policies

PHEFT2 is set with certain build-in attributes called "Policies" that control the PHEFT2 operations. Policies represent the PHEFT2 Appliance Admin's enabling (or disabling) of features that are to apply to the PHEFT2.

The policies can be altered by the EFT Admin as per the requirement.

1. From LunaSH command, login as an admin. Please refer to C.1 Admin User Login LunaSH section.
2. Then login as an EFT Admin. Please refer to the C.3 EFT Admin Login LunaSH section.
3. To view current HSM policies, perform the following command:
   `lunash:>hsm showpolicies`
4. To change a current HSM policy, perform the following command:
   `lunash:>hsm changepolicy -p <hsmpolicynumber> -v <hsmpolicyvalue>`
5. To enable auto hsm activation for future reboot, you need to disable required of activation eToken login, perform the following command:
   `lunash:>hsm changepolicy -p 101 -v 0`
6. Table below provides a brief discription of the purpose and operation of policy name 101.

Policies	Description	Policy Name	Destructive	Modifiable
Activation Token	required of activation eToken login	101	No	Yes

Creating PHEFT2 HSM Partition

HSM Partitions are independent logical HSMs with separate cryptographic work-spaces that reside within the PHEFT2 appliance. Each partition has its own data, access controls and security policies, independent from other partitions. The PHEFT2 currently supports single partition with a special administration account, called the Partition Owner, who manages it. You can create and delete a partition as per your requirements. The default key storage space is 16 MB, out of which 15.5 MB is allotted for storage. Multiple partition support is available from 2.1.0 HSM and above. A maximum of 20 partitions can be created, explicitly licensed, either from the factory, or by upgrade. Hence, the user has the option to use the factory-default single HSM partition to upgrade to 5, 10, 15 or 20 partitions.

To create partition, perform the following steps:

1. From LunaSH command, login as admin. Please refer to the C.1 Admin User Login LunaSH section.
2. Then login as EFT Admin. Please refer to the C.3 EFT Admin Login LunaSH section.
3. Add a partition on the PHEFT2 using the command: `lunash:>sysconfig partition create -partition <name>`
4. Software Version 2.1.0 HSM and above: `lunash:>sysconfig partition create -partition <name> -s <size>` or `lunash:>sysconfig partition create -partition <name> -a <use all free storage space>`

   > **Note**: To specify partition size, check out the free storage space in HSM using the command `hsm show`. Alternatively, simply configure the partition to use all the remaining free storage space in the HSM. If the created partition's size is too small, it may affect key creation.

5. Enter "proceed" to continue partition creation.
6. Enter the username for Partition Owner 1.
7. Insert the black eToken for partition Owner creation. The system prompts you to enter the PIN for the Partition Owner.
8. Re-enter the PIN to confirm.
9. Repeat steps (5 to 7) for Partition Owner 2.
10. On successful initialization, the message "Partition created successfully" is displayed.

Managing HSM RSA Keys

On boot up, 64 RSA key pairs are generated automatically inside the PHEFT2 but only index 1-16 to be used for now. These keys are used for the supported AS2805 processing of the HSM.

1. From LunaSH command, login as admin. Please refer to the C.1 Admin User Login LunaSH section.
2. Then login as EFT Partition Owner. Please refer to the C.4 EFT Partition Owner Login LunaSH section.
3. Perform the following command to view existing HSM RSA key:

   ```
   lunash:>keyMgmt view hsm rsa -i <index>
   ```
4. Perform the following command to generate 16 HSM RSA keys one by one:

   ```
   lunash:>keyMgmt generate hsm rsa -i <index> -l <modulusLength> -e <publicExponent>
   ```

> **Note**:
>
> - The modulus length must be between 512 and 2048 (increments of 64 bits). The default key length for these keys is 1024 bits. It is recommended to use keys with modulus length multiples of 128 bits, as PHEFT2 can perform RSA operations for such moduli lengths faster.
> - The public exponent choices are (3, 65537, RANDOM). Do not use 3.

Managing Host-stored RSA Keys

The PHEFT2 HSM has a table that stores a maximum of 16 key-pairs that are pre-generated on request. Once a key pair is retrieved, it is deleted from the table and a new one is generated.

If a requested key type is not in the table, a key of that type is added to the table and generations starts. If the table is full, the error "Public Key Pair not available" is returned. If a requested key type is in the table but all keys of that type have been used, the error "Public Key Pair generating" is returned.

1. From LunaSH command, login as an admin. Please refer to the C.1 Admin User Login LunaSH section.
2. Then login as an EFT Partition Owner. Please refer to the C.4 EFT Partition Owner Login LunaSH section.
3. Perform the following command to view Public Verification Codes (PVC) for each of the host-stored RSA key pairs:

   ```
   lunash:>keyMgmt view host rsa
   ```
4. Perform the following command to generate 16 HSM RSA keys one by one:

   ```
   lunash:>keyMgmt generate host rsa -n <numberOfkeys> -l <modulusLength> -e
   <publicExponent> -g <keyGroup>
   ```

ℹ	**Notes**: ○ The modulus length must be between 512 and 2048 (increments of 64 bits). The default key length for these keys is 1024 bits. ○ The public exponent choices are (3, 65537, RANDOM). Do not use 3.

Enabling Transferable HSM RSA keys

In order to configure the PHEFT2 HSM to transfer the HSM RSA keys to smartcard in backing up operation, please do the following steps:

1. From Luna EFT Administration Console, login as EFT Partition Owner. Please refer to the C6.2 EFT Partition Owner Login Luna EFT Administration Console section.
2. From the left menu panel, navigate to **Payment Configuration> Key Settings**. The following page is displayed:

3. Click on the check box for **Enable Transfer of HSM RSA Key Pair**.
4. Click the **Save** button to save your setting.

Configuring Security Module Host Comms

Security Module Host Comms Configuration utility is used to administer the communication between AccessMatrix and Security Module like HSM. The connection information of HSM like IP address and port number are required for this configuration. The utility can also be used to set up Failover Load Balancing, as explained in Failover Load Balancing.

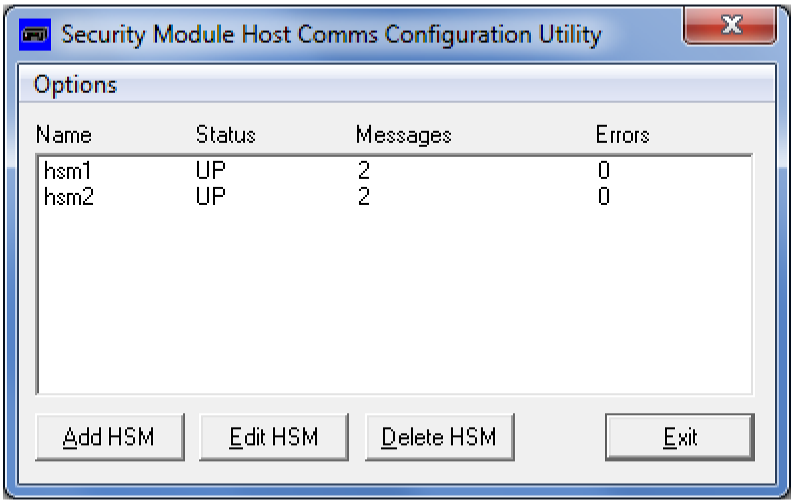
ℹ	**Note**: The utility is contained in the ETptkcomm package.

For each HSM, you can configure the **Name**, **Weight**, **Window**, **Timeout**, and **Heartbeat**, and mark the **HSM Printer attached** and **Stand by only** options.

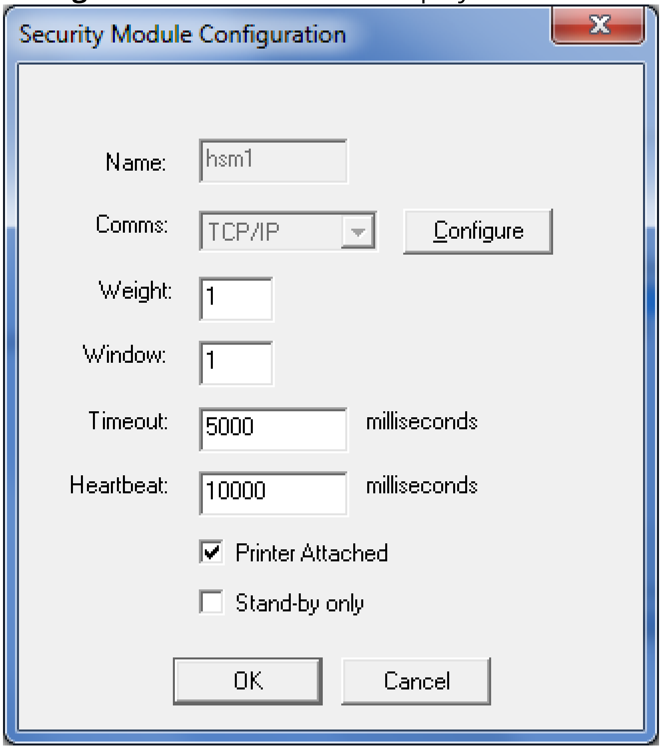
> **Notes**:
>
> - **Name** must be between one to four characters.
> - **Weight** specifies the distribution of the workload among HSM's. For example: If there are two HSM's, HSM1 with weight of four and HSM2 with weight of one, then HSM1 handles four functions before HSM2 handles one function.
>   - **Weight** must be greater than zero in order the HSM to be used. Weight of zero means the HSM is disabled or will not be used.
> - **Window** specifies the number of send messages that can be outstanding to this security module before the receive queue is allowed time to catch up.
>   - If **Window** is greater than one, there is no guarantee that the responses will be returned in order.
> - **Timeout** is the amount of time to wait from sending the message before returning before returning an error or retry the send operation.
> - **Heartbeat** is the amount of time between sending heartbeat messages to the HSM. Heartbeat message is used to update the status of the HSM (UP or DOWN).
>   - **Heartbeat** should be greater than timeout.
> - **Printer attached** specifies that this HSM is connected to a printer for PIN or password printing.
> - **Stand by only** specifies that this HSM is only be used when there is no other HSM available. For HSM with printing task only, this option must be set (checked).

Security Module Configuration using Windows GUI

1. In the AccessMatrix server, run the Security Module Configuration Utility. The **Security Module Host Comms Configuration Utility** window will be displayed, as shown in below.



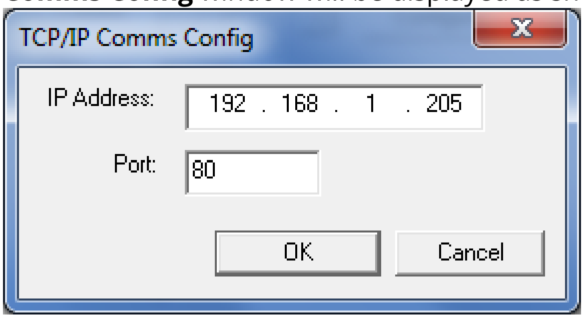
2. Click on the **Add HSM** button to add an HSM to the list. A **Security Module Configuration** window will be displayed as shown below.



3. Enter the name, weight, window, timeout, heartbeat, printer attached, stand-by only information accordingly.

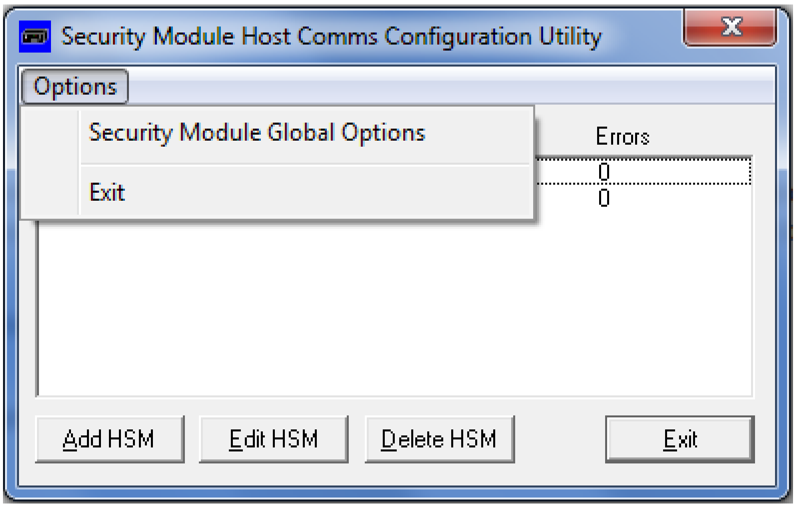
> **Notes**:
> - Please refer to notes in the Configuring Security Module Host Comms section for more details.
> - HSM that handles printing task only, the **Stand by only** option must be set (checked) to prevent this HSM being used for password generation or verification.

4. Select **TCP/IP from Comms** to specify TCP/IP as the communication protocol used to communicate with the HSM.
5. Click on **Configure** to configure the TCP/IP communication with HSM. A **TCP/IP Comms Config** window will be displayed as shown below.

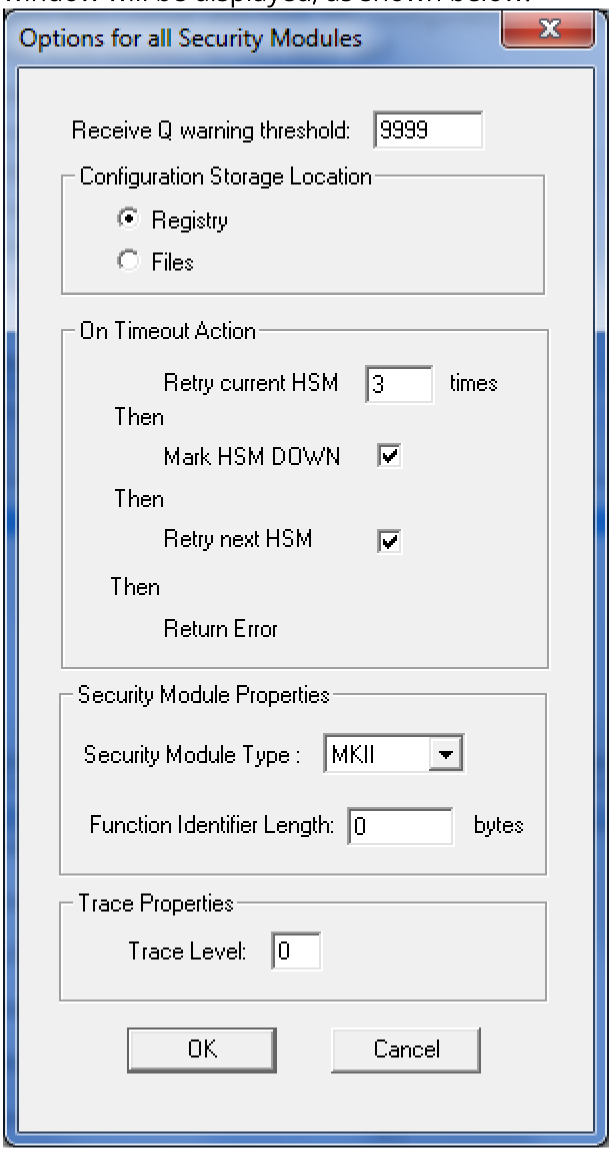


6. Enter the IP address and port of the HSM accordingly.
7. Click on the **OK** button to return to the Security Module Configuration window.
8. Click on the **OK** button to add the HSM to the list of security module and return to **Security Module Host Comms Configuration Utility** window.

9. Click on the **Options** button at the title bar. A context menu will be displayed, as shown below.



10. Click on **Security Module Global Options**. The **Options for all Security Modules** window will be displayed, as shown below.



<table>
<tr><td></td><td>

11. Enter a number for **Retry current HSM** and select the **Mark HSM DOWN** and **Retry next HSM** accordingly.

Note:

- **Retry current HSM** specifies how many times of retry in contacting the HSM after timeout error occurs.
- If **Mark HSM DOWN** is checked, then the status of the timeout HSM will be marked as DOWN after certain number of retries.
- Once HSM has been marked DOWN, the status of the HSM will be updated according to the heartbeat setting in step 3.
- If **Retry next HSM** is checked, then the next step after marking the HSM DOWN is to retry using the next available HSM.
- If all fails, then the timeout error will be returned to the caller.

12. Click on the **OK** button to save the configuration.

</td></tr>
<tr><td>Security Module Configuration using Console</td><td>

Notes:

- Default location for the Safenet ETptkcomm installation in Solaris is */opt/Protecttoolkit Comm*.
- Default location for the Safenet ETptkcomm installation in Windows is *C:\Program Files\SafeNet\Protecttoolkit Comm*.

1. Locate and execute CommsConfig file (**CommsConfig.exe** for Windows) in the Protecttoolkit Comm installation directory. The `Security Module Host Comm` configuration utility main menu will be displayed, as shown below:

```
Security Module Host Comms configuration utility
Copyright (c) 2009 SafeNet Inc
Version 5.02.00
Runtime Version 5.02.00

*** Main Menu ***
A - Add a new Security Module configuration
E - Edit a Security Module configuration
D - Delete a Security Module configuration
P - edit global Security Module options
ESC to exit
```

</td></tr>
</table>

2. Press A on the keyboard to add an HSM. The utility will ask for information about the new HSM configuration.

```
Enter Security Module name : HSM3

name = HSM3
Enter the weight for this Security Module : 1
Enter the window for this Security Module : 10
Enter the timeout (ms) for this Security Module : 3000
Enter the heartbeat interval (ms) for this Security Module : 6000
Is there  printer attached to the Security Module [Y/N] ? : Y
Is this Security Module marked as Standby [Y/N] ? : N

>>Select Security Module connection>>>>
 0 - TCP/IP
 1 - ASYNC
 2 - PCI
 ESC to cancel
 ENTER to accept default (*)

Enter IP address : 192.168.1.20
Enter the port  : 8888
```

3. Enter the name, weight, window, timeout, heartbeat, printer attached, stand-by only information, accordingly.

i	**Note**: Please refer to notes in Configuring Security Module Host Comms section for more details.

4. Choose TCP/IP by entering "0" when selecting the connection method for the new HSM.

5. Enter the IP address and port number of the HSM.

6. From Main Menu, press P on the keyboard button to modify the Global Security Module Options.

```
Edit Globals
Enter the Q Threshold : 8888

>>Select action on Timeout>>>>
*0 - Perform Retries
 1 - Error Immediately
 ESC to cancel
 ENTER to accept default (*)

Enter the number of times to retry the Security Module : 3
Mark Security Module as DOWN on Timeout [Y/N] ? : Y
Try Next Security Module [Y/N] ? : Y

>>Select Security Module functionality>>>>
*0 - MKII
 1 - AMB
 ESC to cancel
 ENTER to accept default (*)
```

7. Enter the global queue threshold.

8. Choose the action taken after HSM timeout.

9. Choose whether to mark the failure HSM DOWN or not after the timeout.

i	**Note**: Once HSM has been marked DOWN, the status of the HSM will be updated according to the heartbeat setting in step 3.

	10. Choose whether to try the next HSM or not after marking the failure HSM DOWN. **Note**: To be able to set this setting, mark the failure HSM DOWN must be set. 11. You will be asked for Security Module Functionality. You do not need to set this option, so press the ESC button on the keyboard. The Main Menu will be displayed again.

Failover Load Balancing

Failover load balancing allows two or more HSM's to share the work, according to the failover load balancing configuration. It also provides higher reliability, as other HSMs can take over the work of a failed HSM.

> **Notes**:
>
> - Security Module Configuration in Safenet ETptkcomm is required for configuring failover load balancing. Please install the Safenet ETptkcomm in AccessMatrix server before continuing. Refer to Configuring Security Module Host Comms.
> - Printing task cannot be shared among several HSM's. Thus, only one HSM should handle the printing task.
> - HSM that handles printing task only, the Stand by only option must be set (checked) to prevent this HSM being used for password generation or verification.
> - Backup and restore necessary keys from the main HSM to new HSM, please refer to Backing Up and Restoring Using SmartCard.

Poweroff or Rebooting PHEFT2 HSM

Poweroff

This is useful when shifting PHEFT2 appliance and needs power-off.

1. From LunaSH command, login as admin. Please refer to C.1 Admin User Login LunaSH section.
2. To poweroff the PHEFT2 HSM, perform the following command:

   `lunash:>hsm poweroff`

Reboot

This is useful if the system is not behaving and needs reboot. Restarts the Luna EFT appliance by gracefully shutting down all running processes.

1. From LunaSH command, login as admin. Please refer to C.1 Admin User Login LunaSH section.

2. To reboot the PHEFT2 HSM, perform the following command:

   `lunash:>hsm reboot`

3.2. Configuring the PHEFT2 HSM Printer

Printer Connection

The printer port interface is configured as a DTE. Most serial printers are also configured as DTEs. To connect the two, cross-over cable and adapter must be used. There is no standard interface for serial printers. PHEFT2 HSM provides the serial cable and the adapter along with the HSM. The serial cable is called Null-Modem Serial Cable and the adapter is called USB to Serial Adapter.

Example of printer connection (may vary for some printers):

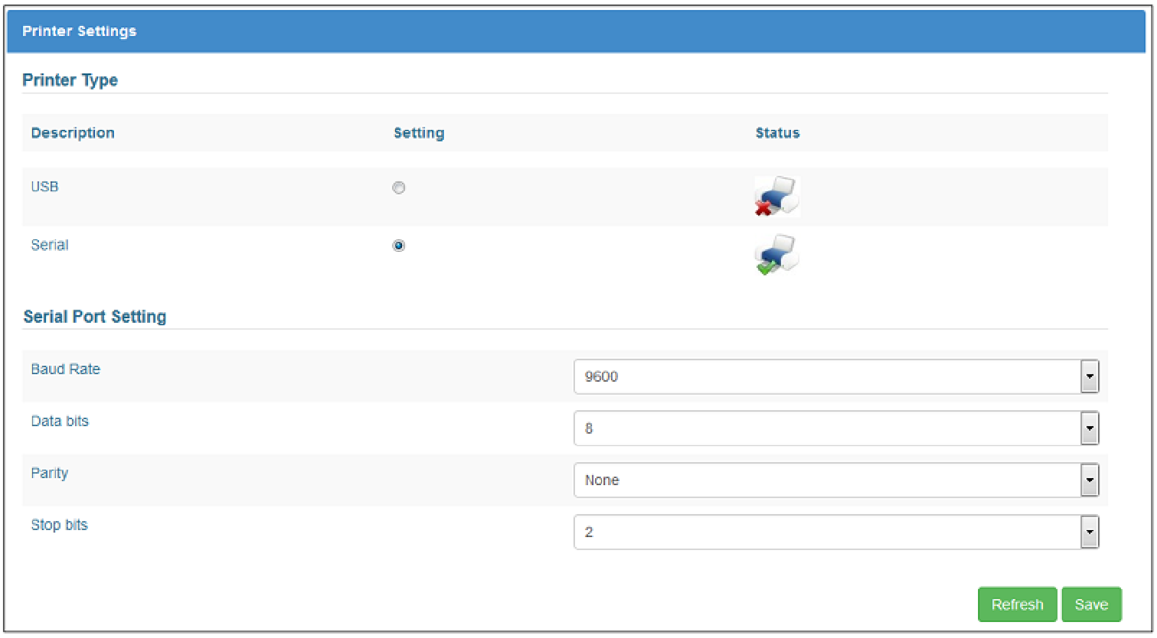
Printer			HSM	
	DTE		DTE	
RX	Pin 3	←	Pin 3	TX
RTS	Pin 4	→ Flow Control Busy →	Pin 8	CTS
DSR	Pin 6	← HSM Present ←	Pin 4	DTR
DTR	Pin 20	→ Printer Operable →	Pin 6	DSR

Configuring Printer Setting

Printer must be properly configured for printing function. Please do the following to configure the printer port:

1. From Luna EFT Administration Console, login as EFT Admin. Please refer to the C.6.1 EFT Admin Login Luna EFT Administration Console section.

2. From the left menu panel, navigate to **System Configuration> Printer Configuration> Printer Setting**. The **Printer Settings** configuration page appears, as shown below.



3. This page allows you to see the printer's status if it is available and connected using USB or Serial connection.

4. Connect printer with HSM and click the **Refresh** button at bottom of the printer setting page to make sure printer status is on.

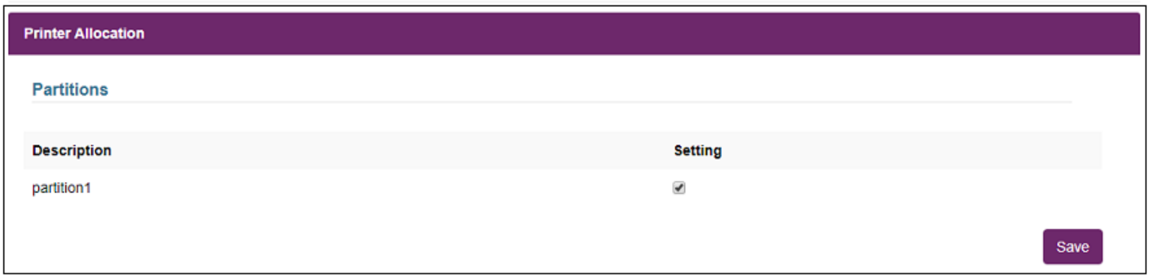
5. Configure **Serial Port Setting** - **Baud Rate**, **Data bits**, **Parity bits** and **Stop bits**.

i	**Note**: The setting may vary from one printer to other printer. Please refer to the printer manual.

6. Click the **Save** button after finishing configuration.

Configure printer allocation

1. In software version 2.1.0 and above, to configure the printer allocation to particular partition, from the left menu panel, navigate to **System Configuration> Printer Configuration> Printer Allocation**. The **Printer Allocation** configuration page appears, as shown below.



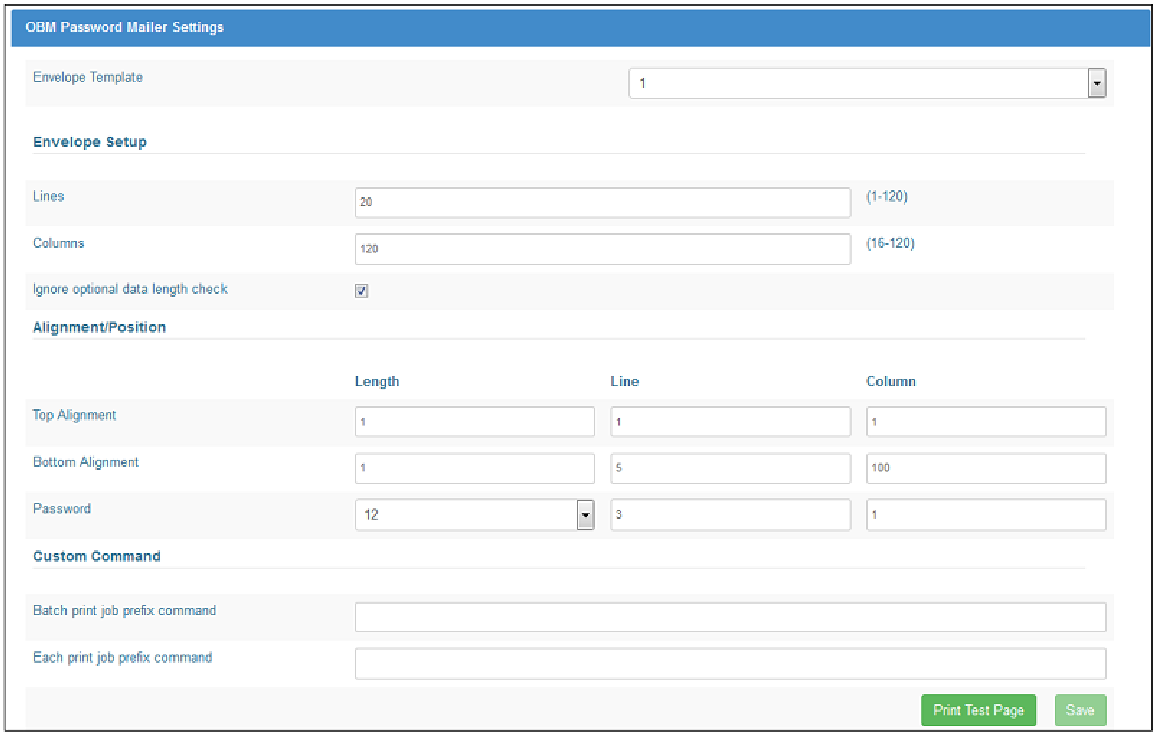
2. All created partition(s) display in this page.

3. By default, printer is not assigned to any partition, and hence, multiple partitions are able to use the printer as a shared resource.
4. Select partition(s) that you wish to prioritize the print job activity as the other partitions will not be able to print or choose the option to perform a print.
5. Click the **Save** button after finish configuration.

Configuring Envelope Template Printing Alignment (Printing Parameters)

In order to print the password on envelope properly, the printing parameters must be properly set to align the printed password on the envelope:

1. From Luna EFT Administration Console, login as the EFT Partition Owner. Please refer to the C.6.2 EFT Partition Owner Login Luna EFT Administration Console section.
2. From the left menu panel, navigate to **Payment Configuration> Envelope Setup> OBM Password Printing**. The **OBM Password Mailer Settings** page appears, as shown below.



3. This page allows you to set at least 10 different envelope templates.
4. Enter the size and location information under the **Envelope Setup** header:
   - **Lines** - Height of the envelope, measured in number of lines.
   - **Columns** - Width of the envelope, measured in number of characters.

- **Ignore optional data length check** - Ignore the envelope column check while printing optional data. This allows user to print data constituting multibyte characters set.

5. The alignment envelope facility prints a series of asterisks to indicate the position of each printed field. There is no other restriction on the position or size of the asterisk fields except that they have to be contained within the envelope frame and cannot overlap with any other console entered parameter field. Configure the following settings under the **Alignment/Position** header as needed:

   - **Top Alignment** - Printing top alignment.
   - **Bottom Alignment** - Printing bottom alignment.
   - **Password** - Password length and the print position. The password may be from 4 to 16 characters in length.

6. The control strings make it possible for the printer to be configured so that it is more suitable for a particular OBM password mail envelope design. For example, the printer may be configured to print a more suitable number of lines per inch. Each control string consists of an even number of hexadecimal characters in the defined range of 01 to FF.

i	**Note**: The control string entries have to be left justified and cannot contain embedded spaces.

   Configure the following settings under the **Custom Command** header as needed:

   - **Batch print job prefix command** - Send the first printer control string to the printer before the first envelope and at the start of the OBM password mailer run.
   - **Each print job prefix command** - Send the second control string to the printer before every envelope (including the first envelope) in a OBM password mail run.

i	**Note**: Please refer to your printer manual for the control string, as different printer has different control string. String sent at start of print run is sent to printer before the first envelope and at the start of password mailer run. String sent at start of each envelope is sent to printer before each envelope.

7. Click the **Print Test Page** button to test the printing of OBM password. After successful print test, then only can save the configuration.

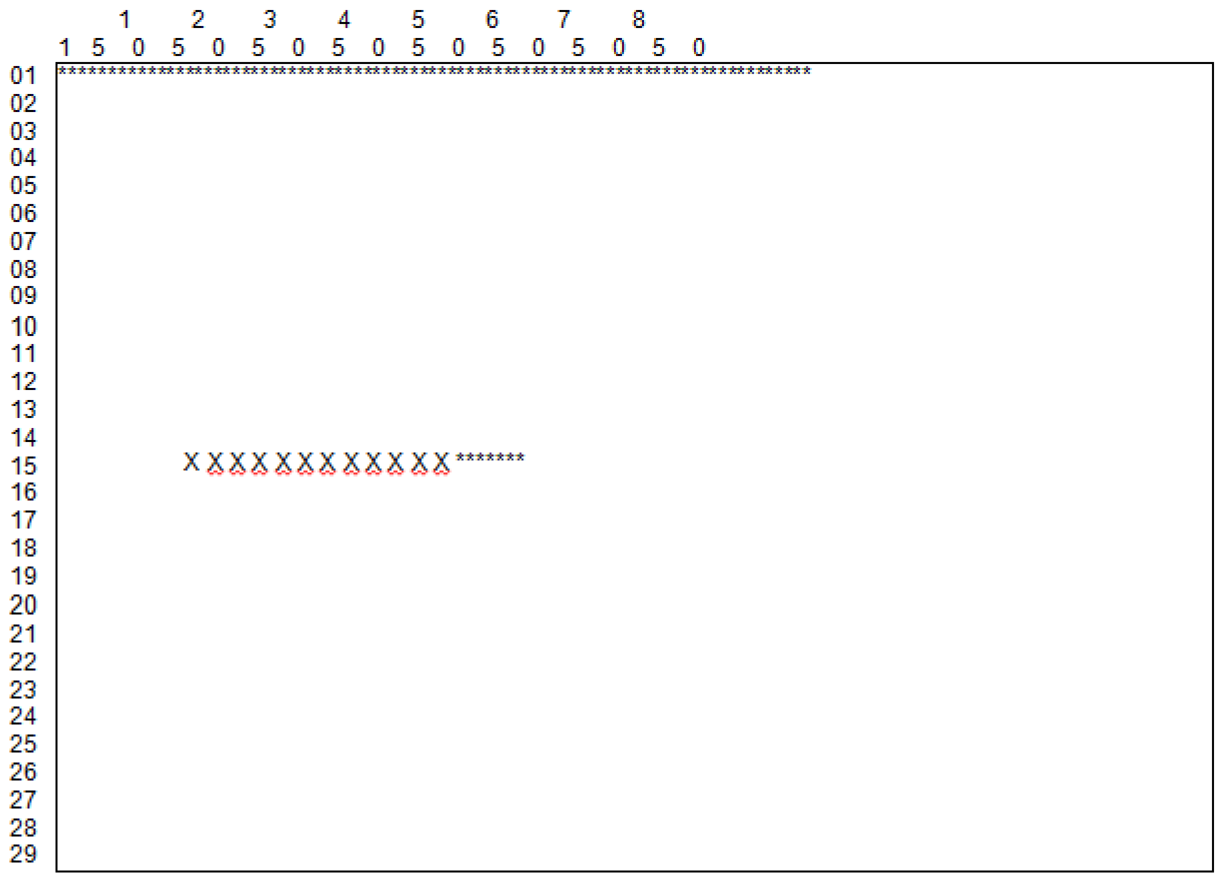
8. Click the **Save** button to store the configuration settings.

Printing Alignment Examples

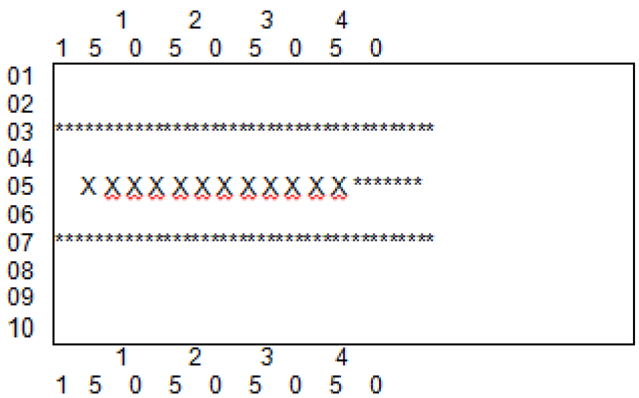
Please enter the following information for Printing Alignment (Printing Parameters):

Alignment Information	Example1			Example2		
Envelope setup - Lines	30			10		
Envelope setup - Columns	80			40		
Envelope setup - Ignore optional data length check	√			√		
	Length	Line	Column	Length	Line	Column
Alignment/Position - Top Alignment	80	1	1	40	3	1
Alignment/Position - Bottom Alignment	80	30	1	40	7	1
Alignment/Position - Password	16	15	20	16	5	5

Below is the result of example 1. The box represents the envelope, as specified in **Envelope setup- Lines** and **Columns**. Asterisks (*) represent the **Alignment/Position** - **Length**, **Lines**, and **Columns**.



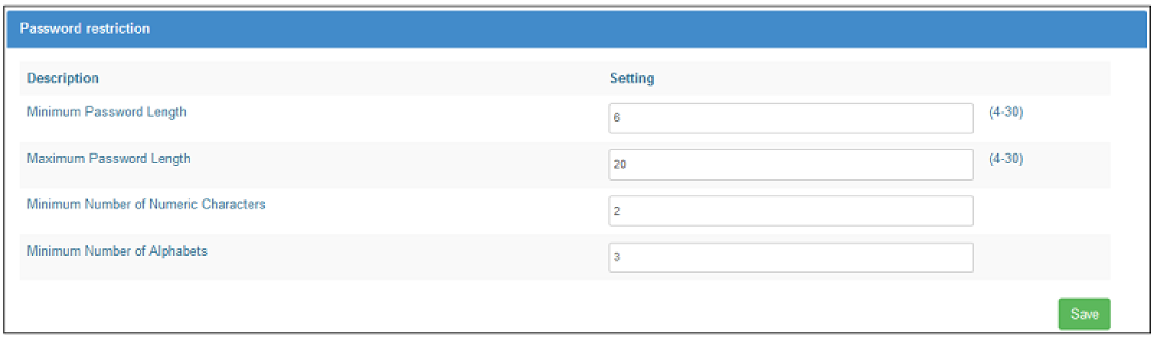
Below is the result of example 2.



Configuring Password Restriction

The Luna EFT Administration Console allows you to configure the rules to be applied for password generation.

1. From Luna EFT Administration Console, login as the EFT Partition Owner. Please refer to the C.6.2 EFT Partition Owner Login Luna EFT Administration Console section.
2. From the left menu panel, navigate to **Payment Configuration> Online Banking Module> Rules**. The **Password Restriction** page appears, as shown below.



3. You can configure the following types of rules:
    - **Minimum password length** - Enter the minimum generated password length in the range of 4 to 30 characters
    - **Maximum password length** - Enter the maximum generated password length in the range of 4 to 30 characters
    - **Minimum number of Numeric Characters** - Enter the minimum numeric characters to be supported in password. This will default to zero, but may be altered (up to the minimum password length) by entering the number required in this field.

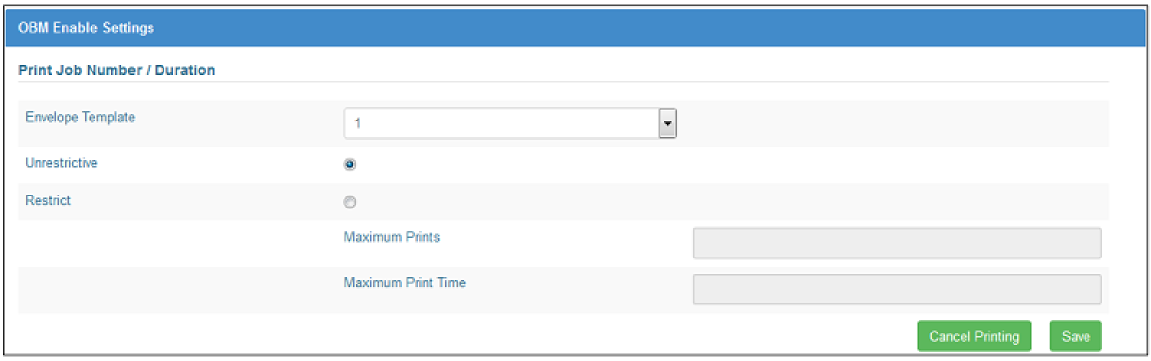
- **Minimum number of Alphabets** - Enter the minimum alphabets to be supported in password. This will default to zero, but may be altered (up to the minimum password length) by entering the number required in this field.

4. Click the **Save** button to store the settings.

Configuring OBM Password Mailer Setting

The Luna EFT Administration Console allows you to control and configure OBM password mailer setting. You can configure the number of password mailers to be printed, and also the time allowed for printing.

1. From Luna EFT Administration Console, login as EFT Partition Owner. Please refer to the C.6.2 EFT Partition Owner Login Luna EFT Administration Console section.
2. From the left menu panel, navigate to **Payment Configuration> Online Banking Module> Settings**. The **OBM Enable Settings** page appears, as shown below.



3. Select the **Envelope Template** (in the range of 1-10) for which you wish to enable the password mailer.
4. Select **Unrestrictive** if you do not want impose any restriction on password mailer.
5. Select **Restrict** to impose a limit on the number of password mailers to be printed, and also on the time allowed for printing. These parameters have to be explicitly entered each time the password mailing console operations are accessed.
   - **Maximum Prints** - Enter the number of passwords that you wish to print.
   - **Maximum Print Time** - Enter the maximum time you wish to allow for the print to occur.
6. Click the **Save** to store the settings.

3.3. Key Processing

Key Processing Terminology

- **Dual Custody** - To assist in preserving key secrecy, at least two custodians are needed for all stored keys. Each has his own secret key component so neither custodian knows the whole key.
- **Clear Key Components** - A clear key component is a secret value known only to one custodian. Depending on the length of the key to which it belongs, it is a 16 or 32, hexadecimal digit number. Each clear key component has a component number associated with it.
- **Encrypted Key Component** - An encrypted key component is similar to a clear key component, but it is treated differently by the key entry mechanism. It also has a component number associated with it.
- **The Resultant Key** - The key components described above are combined by the HSM to form a resultant key. This is the value stored in the HSM. The resultant key is never displayed.
- **Key Verification Codes (KVC)** - A KVC is used to verify, without compromising secrecy, that a key or a key component has been entered correctly, or to confirm that the value of a stored key is as expected. During key entry at the HSM console, a KVC is automatically calculated and displayed for each key component and for the resultant key. Additionally, facilities are provided to display the KVC's of the keys stored in each key table in secure memory.

Storing a Key

A common key storage system is used for all keys. Each key in the PHEFT2 HSM is uniquely identified by its type and its index. Each key type may have specific requirements.

When storing a key you need to know:

- The type of the key.
- The index of the key (its storage location).
- The number of clear key components it is comprised of.
- The number of encrypted key components it is comprised of.

A number of keys of each type can be stored at different index locations within the HSM. The number of index locations available depends on the key type.

Domain Master Keys (KM)

Domain Master Key is used to encrypt other keys in the HSM or application passwords (e.g. database password). The PHEFT2 HSM can store three Domain Master Keys, defined as NEW_KM, CURRENT_KM, and OLD_KM. CURRENT_KM is the Domain Master Key that is currently used for encryption and decryption.

In order to change current Domain Master Key (CURRENT_KM), a new Domain Master Key (NEW_KM) must be entered. When a new Domain Master Key is entered, the key will be stored as NEW_KM. After that, the operator must establish the new Domain Master Key. This will move the CURRENT_KM to OLD_KM and NEW_KM to CURRENT_KM. Then a host function must be called to migrate the host-stored keys encrypted using the old Domain Master Key (OLD_KM) to the current one (CURRENT_KM). After all host-stored keys have been migrated, the OLD_KM can be deleted.

Notes:

- For creation of Domain Master Key, please refer to the Creating Domain Master Key (KM) section.
- For establishing the new Domain Master Key (NEW_KM), please refer to the Establishing New Domain Master Key (NEW_KM) section.
- To enable key migration for host-stored keys, the key migration must be enabled. Please refer to the Enabling/Disabling Key Migration for Domain Master Key section.
- For activation of new Domain Master Key (NEW_KM), please refer to the Activating New Domain Master Key (NEW_KM) section.
- For deletion of old Domain Master Key (OLD_KM), please refer to the Deleting Old Domain Master Key (OLD_KM) section.

Creating Domain Master Key (KM)

The key custodians may enter their keys for KTPV creation. If the key custodians would like their keys to be randomly generated, please refer to the Generating Random Keys or Components section.

Notes:

- The key length is always double-length.
- There is no index number of Domain Master Key.

Please do the following to create a Domain Master Key:

1. From LunaSH command, login as admin. Please refer to the C.1 Admin User Login LunaSH section.
2. Then login as EFT Partition Owner. Please refer to the C.4 EFT Partition Owner Login LunaSH section.
3. Connect a keyboard to HSM.
4. Perform the following command:

   `lunash:>keyMgmt generate hsm km -i <index> -c <clearComp> -e <encryptedComp> -algo <algo> -keyBit <keyBit> -keyLen <keyLength>`

```
[local_host] lunash:>keyMgmt generate hsm km -i 1 -c 2 -e 0 -algo DES -keyLen 3

*********************************************************
*  Enter key component using keyboard attached to HSM  *
*********************************************************

 Enter Clear Key Component#1:****  ****  ****  ****  ****  ****  ****  ****  ****  ****  ****  ****
 KVC: E06D2F
 Please confirm KVC. Do you want to continue? [Y/N]: Y
 Enter Clear Key Component#2:****  ****  ****  ****  ****  ****  ****  ****  ****  ****  ****  ****
 KVC: 4290FA
 Please confirm KVC. Do you want to continue? [Y/N]: Y
 KVC OF KEY: 6611E8
 Please confirm KVC OF KEY. Do you want to Store key? [Y/N]: Y
```

5. Enter clear key component #1 using connected keyboard.
6. A KVC will be generated for confirmation. Enter "Y" to continue.
7. Enter clear key component #2 using connected keyboard.
8. A KVC will be generated for confirmation. Enter "Y" to continue.

> **Notes**:
>
> - Continue if you have more clear components specified.
> - If you specified encrypted component in the command earlier, lunash will prompt you to enter encrypted key component.

9. A KVC of key will be generated for confirmation. Enter "Y" to store the KM.

Establishing New Domain Master Key (NEW_KM)

NEW_KM must be established to move the CURRENT_KM to OLD_KM and NEW_KM to CURRENT_KM.

Please do the following to establish new Domain Master Key (NEW_KM):

1. From LunaSH command, login as admin. Please refer to the C.1 Admin User Login LunaSH section.

2. Then login as EFT Partition Owner. Please refer to the C.4 EFT Partition Owner Login LunaSH section.

3. Connect a keyboard to HSM.

4. Perform the following command:

   `lunash:>keyMgmt generate hsm km -c <clearComp> -e <encryptedComp> -keyLen <keyLength>`

```
[local_host] lunash:>keyMgmt generate hsm newkm -c 2 -e 0 -keyLen 3

**********************************************************
*  Enter key component using keyboard attached to HSM  *
**********************************************************

 Enter Clear Key Component#1:****  ****  ****  ****  ****  ****  ****  ****  ****  ****  ****  ****
 KVC: C391FA
 Please confirm KVC. Do you want to continue? [Y/N]: Y
 Enter Clear Key Component#2:****  ****  ****  ****  ****  ****  ****  ****  ****  ****  ****  ****
 KVC: FAC173
 Please confirm KVC. Do you want to continue? [Y/N]: Y
 KVC OF KEY: 00962B
 Please confirm KVC OF KEY. Do you want to Store key? [Y/N]: Y
```

5. Enter clear key component #1 using connected keyboard.

6. A KVC will be generated for confirmation. Enter "Y" to continue.

7. Enter clear key component #2 using connected keyboard.

8. A KVC will be generated for confirmation. Enter "Y" to continue.

   Notes: Continue if you have more clear components specified.
   If you specified encrypted component in the command earlier, lunash will prompt you to enter encrypted key component.

9. A KVC of key will be generated for confirmation. Enter "Y" to store the newKM.

Enabling/Disabling Key Migration for Domain Master Key

To allow migration of host-stored keys from OLD_KM to CURRENT_KM, the key migration must be enabled. The key migration should be disabled again after all host-stored keys had been migrated to CURRENT_KM.

Please do the following to enable or disable key migration:

1. From LunaSH command, login as admin. Please refer to the C.1 Admin User Login LunaSH section.

2. Then login as EFT Partition Owner. Please refer to the C.4 EFT Partition Owner Login LunaSH section.

3. Perform the following command:

   `lunash:>keyMgmt migrate hsm KM -status <enable/disable>`

```
[local_host] lunash:>keyMgmt migrate hsm KM -status enable

KM migration enabled.
Command Result : 0 (Success)
```

Activating New Domain Master Key (NEW_KM)

To move activated NEW_KM to CURRENT_KM and CURRENT_KM to OLD_KM.

1. From LunaSH command, login as admin. Please refer to the C.1 Admin User Login LunaSH section.

2. Then login as EFT Partition Owner. Please refer to the C.4 EFT Partition Owner Login LunaSH section.

3. To view KM (Current KM) stored at index, run the following command:

   `lunash:>keyMgmt view hsm km -index <index>`

4. To view new KM, run the following command:

   `lunash:>keyMgmt view hsm newkm`

5. Execute the following activateNewKM command to set the newKM as the current KM and the current KM is set to old KM.

   `lunash:>keyMgmt migrate hsm KM -activateNewKM`

```
[local_host] lunash:>keyMgmt migrate hsm KM -activateNewKM

New KM established
Command Result : 0 (Success)
```

6. newKM is now the current KM. To view KM (Current KM) stored at index, run the following command:

   `lunash:>keyMgmt view hsm km -index <index>`

7. newKM is blank. To view new KM, run the following command:

   `lunash:>keyMgmt view hsm newkm`

   1.The current KM is now the old KM. To view old KM, run the following command:

   `lunash:>keyMgmt view hsm oldkm`

Deleting Old Domain Master Key (OLD_KM)

After all host-stored keys have been migrated to CURRENT_KM, the OLD_KM can be deleted.

Please do the following to delete old Domain Master Key (OLD_KM):

1. From LunaSH command, login as admin. Please refer to the C.1 Admin User Login LunaSH section.
2. Then login as EFT Partition Owner. Please refer to the C.4 EFT Partition Owner Login LunaSH section.
3. Perform the following command:

```
lunash:>keyMgmt migrate hsm KM -deletedOldKM
```

```
[local_host] lunash:>keyMgmt migrate hsm KM -deleteOldKM

 RESULT : Old KM erased
 Command Result : 0 (Success)
```

Creating Transformed PIN Value Key (KTPV)

KTPV is used to encrypt passwords. The key custodians may enter their keys for KTPV creation. If want to use HSM to randomly generate key custodians, please refer to the Generating Random Keys or Components section.

Please do the following to create a KTPV:

1. From LunaSH command, login as admin. Please refer to the C.1 Admin User Login LunaSH section.
2. Then login as EFT Partition Owner. Please refer to the C.4 EFT Partition Owner Login LunaSH section.
3. Connect a keyboard to HSM.
4. Perform the following command:

```
lunash:>keyMgmt generate hsm ktpv -i <index> -k <keyLength> -c <clearComp> -e
```

`<encryptedComp>`

```
[local_host] lunash:>keyMgmt generate hsm ktpv -i 1 -k 3 -c 2 -e 0

*********************************************************
*  Enter key component using keyboard attached to HSM  *
*********************************************************

 Enter Clear Key Component#1:****  ****  ****  ****  ****  ****  ****  ****  ****  ****  ****  ****
 KVC: 8CA64D
 Please confirm KVC. Do you want to continue? [Y/N]: Y
 Enter Clear Key Component#2:****  ****  ****  ****  ****  ****  ****  ****  ****  ****  ****  ****
 KVC: 82E136
 Please confirm KVC. Do you want to continue? [Y/N]: Y
 KVC OF KEY: 32W14Q
 Please confirm KVC OF KEY. Do you want to Store key? [Y/N]: Y
```

5. Enter clear key component #1 using connected keyboard.
6. A KVC will be generated for confirmation. Enter "Y" to continue.
7. Enter clear key component #2 using connected keyboard.
8. A KVC will be generated for confirmation. Enter "Y" to continue.

> **Notes**: Continue if you have more clear component specified.
> If you specified encrypted component in the command earlier, lunash will prompt you to enter encrypted key component.

9. A KVC of key will be generated for confirmation. Enter "Y" to store the KTPV.

Creating Data Protect Key (DPK)

Data Protect Key is used for encrypting various application's passwords, e.g. database password, application configuration password. The key custodians may enter their keys for DPK creation. If the key custodians would like their keys to be randomly generated, please refer to the Generating Random Keys or Components section.

Please do the following to create a DPK:

1. From LunaSH command, login as admin. Please refer to the C.1 Admin User Login LunaSH section.
2. Then login as EFT Partition Owner. Please refer to the C.4 EFT Partition Owner Login LunaSH section.
3. Connect a keyboard to HSM.
4. Perform the following command:

`lunash:>keyMgmt generate hsm dpk -i <index> -k <keyLength> -c <clearComp> -e`

<encryptedComp> -a <algo>

```
[local_host] lunash:>keyMgmt generate hsm dpk -i 1 -k 3 -c 2 -e 0 -a DES

*********************************************************
*  Enter key component using keyboard attached to HSM  *
*********************************************************

 Enter Clear Key Component#1:****  ****  ****  ****  ****  ****  ****  ****  ****  ****  ****  ****
 KVC: 8CA64D
 Please confirm KVC. Do you want to continue? [Y/N]: Y
 Enter Clear Key Component#2:****  ****  ****  ****  ****  ****  ****  ****  ****  ****  ****  ****
 KVC: 82E136
 Please confirm KVC. Do you want to continue? [Y/N]: Y
 KVC OF KEY: 73G1S2
 Please confirm KVC OF KEY. Do you want to Store key? [Y/N]: Y
```

5. Enter clear key component #1 using connected keyboard.
6. A KVC will be generated for confirmation. Enter "Y" to continue.
7. Enter clear key component #2 using connected keyboard.
8. A KVC will be generated for confirmation. Enter "Y" to continue.

> **Note**: Continue to if you have more clear components specified.
> If you specified encrypted component in the command earlier, lunash will prompt you to enter encrypted key component.

9. A KVC of key will be generated for confirmation. Enter "Y" to store the DPK.

Creating PIN Protect Key (PPK)

PPK is used for encrypting/decrypting the password before sent for printing after being generated by PIN generation HSM. If the PIN generation and PIN printing functions are separated to two or more HSM's, then all of the HSM's must have the same PPK. For example, if there are three HSM's: HSM A and HSM B for PIN generation, and HSM C for PIN printing only, then all of the HSM's must have the same PPK, so that after PIN generation by HSM A or HSM B, the generated PIN will be encrypted using the PPK before sending it to HSM C.

The key custodians may enter their keys for PPK creation. If the key custodians would like their keys to be randomly generated, please refer to the Generating Random Keys or Components section.

Please do the following to create a PPK:

1. From LunaSH command, login as admin. Please refer to the C.1 Admin User Login LunaSH section.
2. Then login as EFT Partition Owner. Please refer to the C.4 EFT Partition Owner Login LunaSH section.

3. Connect a keyboard to HSM.

4. Perform the following command:

   `lunash:>keyMgmt generate hsm ppk -i <index> -k <keyLength> -c <clearComp> -e <encryptedComp> -a <algo>`

```
[local_host] lunash:>keyMgmt generate hsm ppk -i 1 -k 3 -c 2 -e 0 -a DES

*********************************************************
*  Enter key component using keyboard attached to HSM  *
*********************************************************

 Enter Clear Key Component#1:****  ****  ****  ****  ****  ****  ****  ****  ****  ****  ****  ****
 KVC: 1CA64K
 Please confirm KVC. Do you want to continue? [Y/N]: Y
 Enter Clear Key Component#2:****  ****  ****  ****  ****  ****  ****  ****  ****  ****  ****  ****
 KVC: 52E192
 Please confirm KVC. Do you want to continue? [Y/N]: Y
 KVC OF KEY: 63G187
 Please confirm KVC OF KEY. Do you want to Store key? [Y/N]: Y
```

5. Enter clear key component #1 using connected keyboard.

6. A KVC will be generated for confirmation. Enter "Y" to continue.

7. Enter clear key component #2 using connected keyboard.

8. A KVC will be generated for confirmation. Enter "Y" to continue.

> **Note**: Continue if you have more clear components specified.
> If you specified encrypted component in the command earlier, lunash will prompt you to enter encrypted key component.

9. A KVC of key will be generated for confirmation. Enter "Y" to store the PPK.

Generating Random Keys or Components

PHEFT2 HSM can randomly generate keys or components for key entries, such as KTPV. The generated value was not stored, you need to copy and keep the value. Please follow the following steps for the random generation:

1. From LunaSH command, login as admin. Please refer to the C.1 Admin User Login LunaSH section.

2. Then login as EFT Partition Owner. Please refer to the C.4 EFT Partition Owner Login LunaSH section.

3. Perform the following command:

`lunash:>keyMgmt generate host random -k <keyLength>`

```
[local_host] lunash:>keyMgmt generate host random -keyLen 3

 Triple-length  :       75F7 8CF1 6D2C EA43 4C40 8002 0710 237C E5CD 2CA7 62D6 5E4F
 AS2805.6.3 KVC :       449646

Command Result : 0 (Success)
```

3.4. Backing Up and Restoring Using SmartCard

PHEFT2 HSM provides backup and restore functionalities to save and restore important keys stored in the HSM. The backup procedure can be done by transferring the data to smartcard. The data in the smartcard is encrypted using Smartcard Transfer Protect Key (KTP) and protected by ID and pin. Before restoring the data, the KTP must be entered to the HSM to where the data to be restored in order to decrypt the data. ID and pin are also required to restore the data.

> **Notes**:
> - By default, HSM RSA keys are not transferred to the smartcard. In order to enable the HSM RSA key transfer, please refer to the Enabling Transferable HSM RSA keys section.
> - Backup and restore procedure can be used to copy the important keys to other HSM for failover load balancing, which is explained in the Failover Load Balancing section.
> - The backup may require more than one smartcard to store the data, especially if HSM RSA keys are also to be transferred.
>
> **Warning:**
> The same KTP used to decrypt when restoring from the smartcard must be the same as the KTP used to encrypt the backup data to the smartcard. Thus, it is imperative that KTP must also be stored in a safe place for data retrieval.

SmartCard Transfer Protect Key (KTP)

Before you can transfer keys to, or retrieve keys from the smartcard, you have to enter and store a double length Transfer Protect Key (KTP) using the Store key - KTP console operation. This key is used to derive double length keys to protect all data (including keys) in the smartcard transfer key and receive key process. The KTP is entered in the standard manner as other PHEFT2 HSM keys.

> **Notes**:
> - Please refer to the Key Processing section for more information about key processing.
> - The same KTP used to decrypt when restoring from the smartcard must be the same as the KTP used to encrypt the backup data to the smartcard.

Please do the following to store KTP:

1. From LunaSH command, login as admin. Please refer to the C.1 Admin User Login LunaSH section.

2. Then login as EFT Partition Owner. Please refer to the C.4 EFT Partition Owner Login LunaSH section.

3. Connect a keyboard to HSM.

4. Perform the following command:

```
lunash:> keyMgmt generate hsm ktp -i <index> -c <clearComp> -e <encrpyptedComp> -a <algo> -KeyB <keyBit> -KeyL <keyLength>
```

> **Note**: If algo is DES, do not need to specify keyBit.

```
[local_host] lunash:>keyMgmt generate hsm ktp -i 1 -c 2 -e 0 -a DES -keyL 3

************************************************************
*  Enter key component using keyboard attached to HSM  *
************************************************************

 Enter Clear Key Component#1:****  ****  ****  ****  ****  ****  ****  ****  ****  ****  ****  ****
 KVC: 8CA64D
 Please confirm KVC. Do you want to continue? [Y/N]: Y
 Enter Clear Key Component#2:****  ****  ****  ****  ****  ****  ****  ****  ****  ****  ****  ****
 KVC: 82E136
 Please confirm KVC. Do you want to continue? [Y/N]: Y
 KVC OF KEY: 72R619
 Please confirm KVC OF KEY. Do you want to Store key? [Y/N]: Y

Command Result : 0 (Success)
[local_host] lunash:>
```

5. Enter clear key component #1 using connected keyboard.

6. A KVC will be generated for confirmation. Enter "Y" to continue.

7. Enter clear key component #2 using connected keyboard.

8. A KVC will be generated for confirmation. Enter "Y" to continue.

> **Notes**: Continue to insert if you have more clear component specified. If you specified encrypted component in the command earlier, lunash will prompt you to enter encrypted key component.

9. A KVC of key will be generated for confirmation. Enter "Y" to store the KTP.

Storing or Backing up Keys to SmartCard

> **Note**: The same KTP used to decrypt when restoring from the smartcard must be the same as the KTP used to encrypt the backup data to the smartcard.

1. From LunaSH command, login as admin. Please refer to the C.1 Admin User Login LunaSH section.

2. Then login as EFT Partition Owner. Please refer to the C.4 EFT Partition Owner Login LunaSH section.

3. Perform the following command:

   `lunash:> support backup smartcard -c <cardsetid>`

```
[local_host] lunash:>support backup smartcard -cardsetid isprint

Please insert smart card and retry.

Enter user pin:: *******
Confirm user pin::*******

Insert Card (1) and press(Y/N):Y

Checking Card...
This card may contain data.
Existing card data will be destroyed.
Do you want to continue(Y/N)?Y

Erasing card...

Key transfer is in progress...
```

4. A message will be displayed to ask you insert smartcard. Connect card reader to HSM and insert a smartcard.

5. Enter user pin.

6. Insert empty card for keys backup, if card contains any data, existing card data will be overwritten by new data.

7. Enter "Y" to continue backup process.

8. After completed transfer data to one card, the total transferred byte will be displayed.

   Note: lunash will required more card if needed.

9. At the end of the data transfer, lunash will ask you to insert the first card again for writing the total number of cards used. Please do so.

10. A message will appears to let you know that the keys successfully loaded.

Retrieving or Restoring Keys from SmartCard

Notes:

- The same KTP used to decrypt when restoring from the smartcard must be the same as the KTP used to encrypt the backup data to the smartcard.
- Existing key in the HSM will be replaced if the existing key has the same index number as the one in the smartcard. However, if the

keys in the smartcard do not use the existing key's index, then the existing key will not be deleted or replaced.

- If the smartcard contains RSA keys, the RSA keys will also be transferred or restored from the smartcard.

1. From LunaSH command, login as admin. Please refer to the C.1 Admin User Login LunaSH section.
2. Then login as EFT Partition Owner. Please refer to the C.4 EFT Partition Owner Login LunaSH section.
3. Perform the following command:

   `lunash:> support restore smartcard -c <cardsetid>`

```
[local_host] lunash:>support restore smartcard -cardsetid isprint


NOTICE:HSM keys will be overwritten.
If you are sure that you wish to restore smart card data, then type 'pro
ceed', otherwise type 'quit'

> proceed
Proceeding...

Enter User pin::*******

Please insert smart card and retry.
Reading from card ...

Key transfer in progress...
Bytes Transferred : 33840

Smart card key transfer is completed.
```

4. A notice displayed to warn you HSM keys will be overwritten, type "proceed" to continue.
5. Enter user pin of the smartcard.
6. A message prompt you to insert card. Insert the smartcard which contained backup keys that you wish to restore.
7. At the end of the data restore, The total bytes have been restored into HSM will be displayed together with a message indicate that the retrieval has been completed.

4. Appendices

This section contains the following appendices:

- A. Initial Setup Information
- B. Key Component Sample Form
- C. Users and Roles Login
- D. Configuring Secure Communication Channel (Optional)
- E. Erasing PHEFT2 HSM's Memory
- F. Destructive Actions
- G. SafeNet Authentication Client (SAC)

4.1. A. Initial Setup Information

<table>
<tr><th></th><th>Configuration</th><th>Attribute</th><th>Default/Recommended Value</th><th>Value</th></tr>
<tr><td>1.</td><td>HSM RSA key</td><td>Modulus length</td><td>1024</td><td></td></tr>
<tr><td rowspan="4">2.</td><td rowspan="4">Domain Master Key (KM)</td><td>Number of clear components</td><td>2</td><td></td></tr>
<tr><td>Number of encrypted components</td><td>0</td><td></td></tr>
<tr><td>Clear key components</td><td></td><td>You may write the key component on the key component form</td></tr>
<tr><td>Encrypted key components</td><td>[N/A, if number of encrypted is 0]</td><td></td></tr>
<tr><td rowspan="4">3.</td><td rowspan="4">SmartCard Transfer Key (KTP)</td><td>Number of clear components</td><td>2</td><td></td></tr>
<tr><td>Number of encrypted components</td><td>0</td><td></td></tr>
<tr><td>Clear key components</td><td></td><td>You may write the key component on the key component form</td></tr>
<tr><td>Encrypted key components</td><td>[N/A, if number of encrypted is 0]</td><td></td></tr>
<tr><td rowspan="2">4.</td><td rowspan="2">Retrieve all stored keys from Smartcards</td><td>PIN number</td><td></td><td>Do not write the password here</td></tr>
<tr><td>Set ID</td><td></td><td></td></tr>
<tr><td rowspan="2">5.</td><td rowspan="2">Transformed PIN Key (KTPV)</td><td>Key index</td><td></td><td></td></tr>
<tr><td>Number of clear components</td><td>2</td><td></td></tr>
</table>

	Configuration	Attribute	Default/Recommended Value	Value
		Number of encrypted components	0	
		Key length	Double length	
		Clear key components		You may write the key component on the key component form
		Encrypted key components	[N/A, if number of encrypted is 0]	
6.	Data Protect Key (DPK)	Key index		
		Number of clear components	2	
		Number of encrypted components	0	
		Key length	Double length	
		Algorithm	DES	
		Clear key components		You may write the key component on the key component form
		Encrypted key components	[N/A, if number of encrypted is 0]	
7.	PIN Protect Key (PPK)	Key index		
		Number of clear components	2	
		Number of encrypted components	0	
		Key length	Double length	

	Configuration	Attribute	Default/Recommended Value	Value
		Algorithm	DES	
		Clear key components		You may write the key component on the key component form
		Encrypted key components	[N/A, if number of encrypted is 0]	
8.	Printer configuration	Data rate	9600	
		Data bits	8	
		Parity	None	
		Stop bits	1	
9.	Print parameters or alignment	Envelope setup-Lines	30	
		Envelope setup-Columns	80	
		Envelope setup-Ignore optional data length check	Enabled	
		Alignment/Position-Top Alignment	80 & 1 & 1	
		Alignment/Position-Bottom Alignment	80 & 30 & 1	
		Alignment/Position-Password	16 & 15 & 20	
10.	Printer control string entry	String sent at start of print run	[blank]	
		String sent at start of each envelope	[blank]	
11.	Password restriction entry	Minimum password length	6	
		Maximum password length	10	

	Configuration	Attribute	Default/Recommended Value	Value
		Minimum numeric characters	1	
		Minimum alphabetic characters	1	
12.	OBM password mailer	Restrict/Unrestrictive	Unrestrictive	
		Maximum prints		
		Maximum printing time		

4.2. B. Key Component Sample Form

The following page will contain some key component sample form, which can be used for backup or safekeeping. However, key component is a very sensitive information. Thus, if key component is written to the form. The form must be kept safe.

The key component sample forms provided in this guide:

- Domain Master Key (KM)
- Transformed PIN Value Key (KTPV)
- SmartCard Transfer Protect Key (KTP)
- Data Protect Key (DPK)
- PIN Protect Key (PPK)

The sample form can also be used as a template to create a more suitable form for your company and purposes.

i-Sprint Innovations Pte Ltd
Blk 750D Lobby 1 Chai Chee Road
#08-01 TechnoPark @ Chai Chee
Singapore 469004

Telephone: (65) 6244-3900
Facsimile: (65) 6244-8900
Website: www.i-Sprint.com

Online Banking Module (OBM)
Key Type ______________________ – Index ___ – Clear Component ___ of ___

Name of Key Custodian

Clear Component

Key Verification Check (KVC)

4.3. C. Users and Roles Login

C.1 Admin User Login LunaSH

1. When a successful connection is made, a terminal window opens and the prompt "login as:" appears.
2. Type "admin" and press Enter.
3. You are prompted for the admin password. Key in the admin password and press Enter.

```
login as: admin
admin@172.25.13.143's password:

Last login: Fri Dec  12 00:02:33 2014

Luna EFT 2.0 Command Line Shell - Copyright (c) 2014 onwards SafeNet, Inc. All rights reserved.

[local_host] lunash:>
```

C.2 Audit User Login LunaSH

1. When a successful connection is made, a terminal window opens and the prompt "login as:" appears.
2. Type "audit" and press Enter.
3. You are prompted for the audit password. Key in the audit password and press Enter.

```
login as: audit
audit@172.25.13.143's password:

Last login: Fri Dec  12 00:02:33 2014

Luna EFT 2.0 Command Line Shell - Copyright (c) 2014 onwards SafeNet, Inc. All rights reserved.

[local_host] lunash:>
```

C.3 EFT Admin Login LunaSH

Software version below 2.1.0	1. Type the following command: `lunash:> login eftAdmin` 2. You are prompted for the username of the first user. Type the first EFT Admin's name and press Enter. 3. You are prompted to insert the eToken and PIN for the first EFT Admin. Insert the first HSM Admin eToken, key in the PIN and press Enter. 4. An authentication failed message appears in the event of a failure. If successful, you are prompted for username for second user. Type in the second EFT Admin's name and press Enter. 5. You are prompted to insert the eToken and PIN for second EFT Admin. Insert the second HSM Admin eToken, key in the PIN and press Enter.

<table>
<tr><td></td><td>

6. Messages appear to show that the EFT administrator has been authenticated and the login is successful.

```
[local_host] lunash:>login eftAdmin

Enter username for first user : EFTAdmin1
Insert token and enter PIN :
> ********

Enter username for second user : EFTAdmin2
Insert token and enter PIN :
> ********

Authentication Successful.

EFT administrator login successful.
```

</td></tr>
<tr><td>Software version 2.1.0 and above</td><td>

1. You can choose to login with an OTP using the following command:
`lunash:> login eftAdmin -otp`

Note: The Safenet Payment HSM OTP Generator is used to generate OTP. Please refer to the steps on using the Safenet Payment HSM OTP Generator in C.6 Luna EFT Administration Console.

2. You are prompted for the username of the first user. Type in the first EFT Admin's name and press Enter.
3. You are prompted to enter the OTP for the first EFT Admin. Get the generated OTP from Safenet Payment HSM OTP Generator.
4. An authentication failed message appears in the event of a failure. Otherwise, you are prompted for the username for the second user. Type in the second EFT Admin's name and press Enter.
5. You are prompted to enter the OTP for the second EFT Admin. Get the generated OTP from the Safenet Payment HSM OTP Generator.
6. Messages appear to show that the EFT administrator has been authenticated and the login is successful.

</td></tr>
</table>

C.4 EFT Partition Owner Login LunaSH

<table>
<tr><td>Software version below 2.1.0</td><td>

1. Type the following command:
`login partitionOwner -partition <name>`
2. You are prompted for the username of the first user. Type in the first EFT Partition Owner's name and press Enter.
3. You are prompted to insert the eToken and PIN for the first EFT Partition Owner. Insert the first HSM Partition Owner's eToken, key in the PIN and press Enter.
4. An authentication failed message appears in the event of a failure. Otherwise, you are prompted for the username of the second user. Type in the second EFT Partition Owner's name and press Enter.
5. You are prompted to insert the eToken and PIN for the second EFT Partition Owner. Insert the second HSM Partition Owner's eToken, key in the PIN and press Enter.

</td></tr>
</table>

<table>
<tr><td></td><td>

6. Messages appear to show that the EFT Partition Owners have been authenticated and the login is successful.

```
[local_host] lunash:>login partitionOwner -partition part1

Enter username for first user : PartitionOwner1
Insert token and enter PIN :
> ********

Enter username for second user : PartitionOwner2
Insert token and enter PIN :
> ********

Authentication Successful.

Partition owner login successful.
```

</td></tr>
<tr><td>Software version 2.1.0 and above</td><td>

1. You can choose to login with OTP. Safenet Payment HSM OTP Generator is used to generate OTP. Please refer to Safenet Payment HSM OTP Generator in C.6 Luna EFT Administration Console section:
`lunash:> login partitionOwner -partition <name> -otp`
2. You are prompted for the username of the first user. Type in the first EFT Partition Owner's name and press Enter.
3. You are prompted to enter the OTP for the first EFT Partition Owner. Get the generated OTP from the Safenet Payment HSM OTP Generator.
4. An authentication failed message appears in the event of a failure. Otherwise, you are prompted for the username of the second user. Type in the second EFT Partition Owner's name and press Enter.
5. You are prompted to enter the OTP for the second EFT Partition Owner. Get the generated OTP from the Safenet Payment HSM OTP Generator.
6. Messages appear to show that the EFT Partition Owners have been authenticated and the login is successful.

</td></tr>
</table>

C.5 EFT Auditor Login LunaSH

<table>
<tr><td>Software version below 2.1.0</td><td>

1. Type in the following command:
`login eftAuditor`
2. You are prompted for the username of the first user. Type in the first EFT Auditor's name and press Enter.
3. You are prompted to insert the eToken and PIN for the first EFT Auditor. Insert the first HSM Auditor's eToken, key in the PIN and press Enter.
4. An authentication failed message appears in the event of a failure. Otherwise, you are prompted for the username of the second user. Type in the second EFT Auditor's name and press Enter.
5. You are prompted to insert the eToken and PIN for second EFT Auditor. Insert the second HSM Auditor's eToken, key in the PIN and press Enter.

</td></tr>
</table>

	6. Messages appear to show that the EFT Auditor has been authenticated and the login in successful.
	`[local_host] lunash:>login eftAuditor`
	
	`Enter username for first user : audit1`
	`Insert token and enter PIN :`
	`> ********`
	
	`Enter username for second user : audit2`
	`Insert token and enter PIN :`
	`> ********`
	
	`Authentication successful.`
	
	`EFT Auditor login successful.`

Software version 2.1.0 and above	1. You can choose to login with an OTP: `lunash:> login eftAuditor -otp` **Note**: The Safenet Payment HSM OTP Generator is used to generate an OTP. Please refer to the steps on using the Safenet Payment HSM OTP Generator in C.6 Luna EFT Administration Console. 2. You are prompted for the username of the first user. Type in the first EFT Auditor's name and press Enter. 3. You are prompted to enter the OTP for the first EFT Auditor. Get the generated OTP from the Safenet Payment HSM OTP Generator. 4. An authentication failed message appears in the event of a failure. Otherwise, you are prompted for the username of the second user. Type in the second EFT Auditor's name and press Enter. 5. You are prompted to enter the OTP for the second EFT Auditor. Get the generated OTP from the Safenet Payment HSM OTP Generator. 6. Messages appear to show that the EFT Auditors have been authenticated and the login is successful.

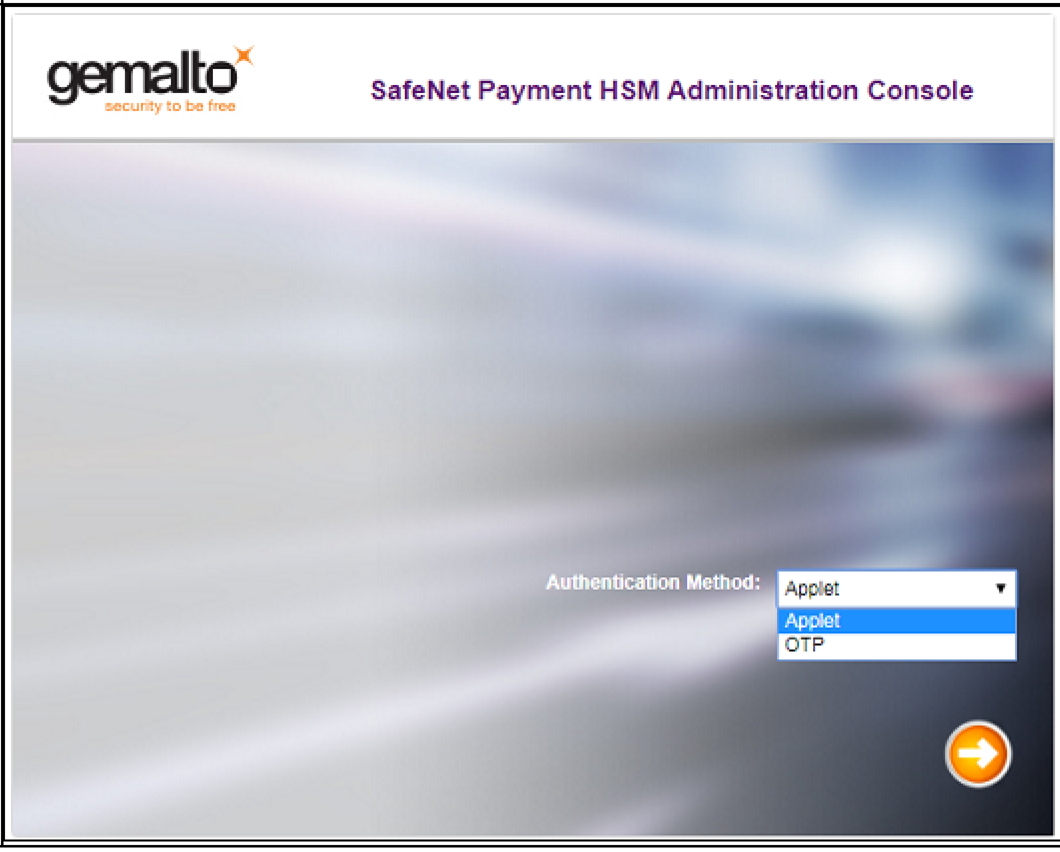
C.6 Luna EFT Administration Console

1. Type the following URL, using the IP address and port you set while establishing host communication:

   https://IP address:<configured port>/eftweb (the default port is 443)

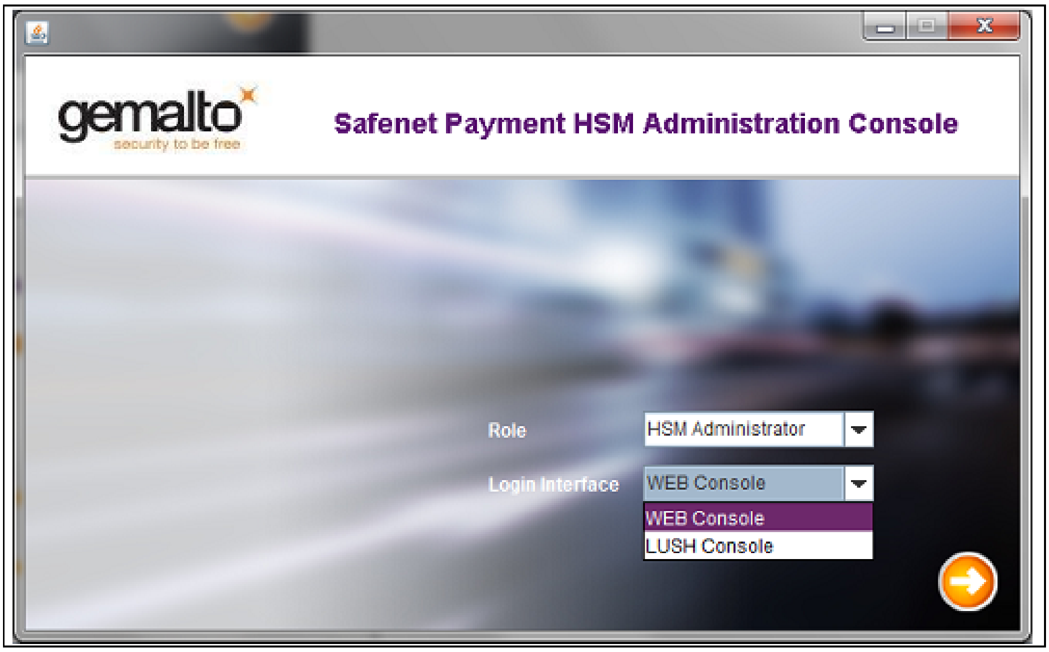
   If using Internet Explorer, enable TLS 1.2 ciphers under “Internet Options”.

   The Luna EFT Administration Console Interface is displayed:

Software version below 2.1.0	
Software version 2.1.0 and above	

2. In addition, major browser distributions are planning to remove support for the Java plugin (applet) due to inherent vulnerabilities. Hence, the OTP is an alternate method to authenticate users logging in to the SafeNet Payment HSM Administration Console interface.
3. Configure **Authentication Method** to "Applet" login (Refer to Software version below 2.1.0 in next subsections) or "OTP" login (Refer to Software version 2.1.0 and above in next subsections).

a. If you choose to use OTP login, download the Safenet Payment HSM OTP Generator from the Luna EFT Administration Console. The Safenet Payment HSM OTP Generator interface is shown below:



b. Select the **Role** "HSM Administrator"/"Partition Owner" and configure **Login Interface** to "WEB Console" to log in as an EFT Admin or EFT Partition Owner in the Web Admin Console.

c. Select the **Role** "HSM Administrator"/"Partition Owner"/"Auditor" and configure the **Login Interface** to "LUSH Console" to log in as an Admin or Partition Owner or Auditor in the LUSH Console.

C.6.1 EFT Admin Login Luna EFT Administration Console

Software version below 2.1.0	1. Select the **Role** "EFT Administrator". 2. Click **Login**. 3. Insert the eToken for the first user and enter the username and PIN. 4. Click **Verify eToken**. 5. Repeat steps 4 and 5 for the second user.
Software version 2.1.0 and above	1. Select the **Authentication Method** "OTP". 2. Click the **Next arrow**. 3. Select the **Role** "HSM Administrator". 4. Click the **Next arrow**. 5. Use the downloaded Safenet Payment HSM OTP Generator to generate OTP for first user. In the Safenet Payment HSM OTP Generator:

	a. Select **Role** "HSM Administrator" and the **Login Interface** "WEB Console". b. Click the **Next arrow**. c. Insert the eToken for the first user and enter the username and PIN. d. The generated OTP is displayed. e. Click **Copy**, followed by the **Done** button. 6. Enter the first username and enter the OTP in the main login screen. 7. Again in Safenet Payment HSM OTP Generator: a. Insert next eToken for the second user and enter the username and PIN. b. A box pop out with generated OTP. c. Click **Copy**, followed by the **Done** button. 8. Enter the second username and enter the OTP in main login screen. 9. Click the **Next arrow** to login.

C.6.2 EFT Partition Owner Login Luna EFT Administration Console

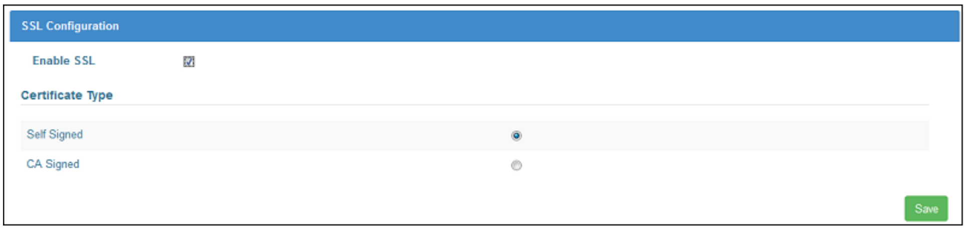
Software version below 2.1.0	1. Select the user **Role** "Partition Owner". 2. Click **Login**. 3. Insert the eToken for the first user and enter the username and PIN. 4. Click **Verify eToken**. 5. Repeat steps 4 and 5 for the second user.
Software version 2.1.0 and above	1. Select the **Authentication Method** "OTP". 2. Click the **Next arrow**. 3. Select the user **Role** "Partition Owner". 4. Enter the **Partition Name**. 5. Click the **Next arrow**. 6. Use the downloaded Safenet Payment HSM OTP Generator to generate an OTP for the first user. In the Safenet Payment HSM OTP Generator: a. Select the **Role** "Partition Owner" and the **Login Interface** "WEB Console". b. Enter the **Partition Name**. c. Click the **Next arrow**. d. Insert the eToken for the first user and enter the username and PIN. e. The generated OTP is displayed. f. Click **Copy**, followed by the **Done** button. 7. Enter the first username and enter the OTP in the main login screen. 8. Again, in Safenet Payment HSM OTP Generator:

	a. Insert next eToken for the second user and enter the username and PIN. b. The generated OTP is displayed. c. Click **Copy**, followed by the **Done** button. 9. Enter the second username and enter the OTP in the main login screen. 10. Click the **Next arrow** to login.

4.4. D. Configuring Secure Communication Channel (Optional)

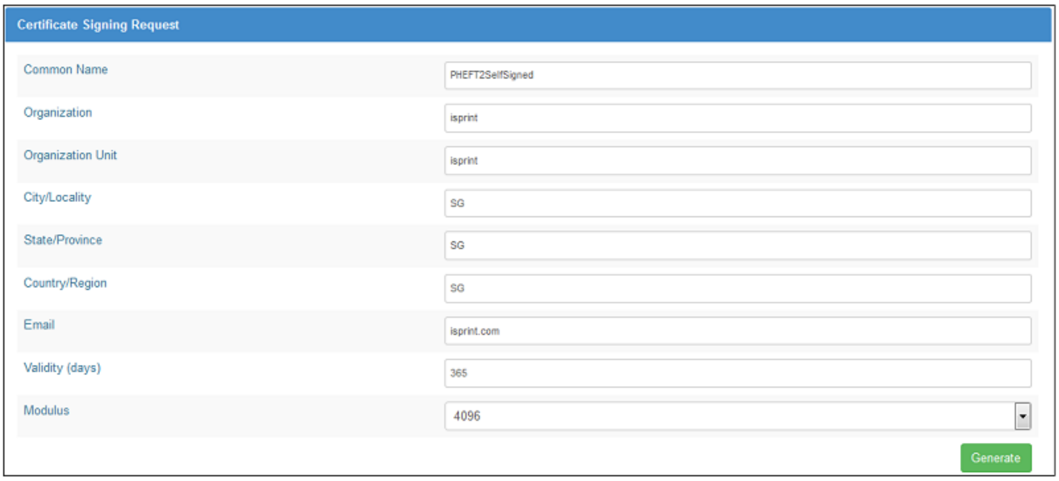
A secure channel builds both an authenticated and an encrypted channel between host and Luna EFT. This would in entirety provide authentication, message authentication and message privacy.

1. In the Luna EFT Administration Console, login as EFT Partition Owner. Please refer to C.6.2 EFT Partition Owner Login Luna EFT Administration Console.
2. To enable and configure SSL communication between PHEFT2 and the host computer:
    a. Go to **System Configuration > Network Configuration > SSL** and tick the **Enable SSL** checkbox for a secure channel. By default, this setting is disabled.
    b. Select the type of certificate:
        - **Self-signed certificate** - Enables the user to register host self-signed certificate with PHEFT2.
        - **CA signed certificate** - Enables the user to register CA signed certificates with PHEFT2.



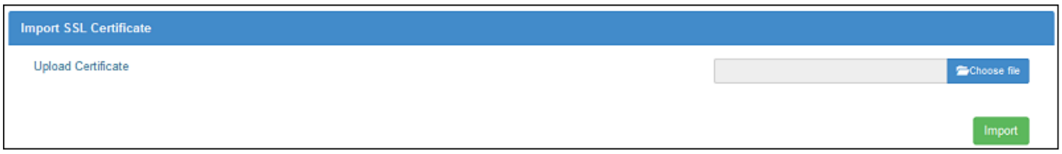
3. To generate CSR and certificate for host communication:
    a. Go to **System Configuration > Certificate Management > Certificate Request** to generate the CSR.
    b. Depending upon the SSL mode, a CSR or self-signed certificate is generated. You can export the CSR, get it signed by a CA, and import it back to Luna EFT to be used as a

PHEFT2 certificate for host communication.



c. Go to **System Configuration > Certificate Management > Export** to export SSL certificate stored on PHEFT2 to the client machine.

d. Go to **System Configuration > Certificate Management > Import** to import SSL certificate on PHEFT2.



4. To register the CA certificate with PHEFT2:

a. Go to **System Configuration > Certificate Management > Manage** and register the certificate.

b. A self-signed certificate is by default registered. You need to register a CA certificate, and CA signed certificate that was imported into Luna EFT.

Notes:

- Certificate once registered for a particular Server/Host/CA type, can not be changed. You will have to delete the certificate and then register the certificate.

	• Due to PCI HSM restrictions, certificates with MD5 and SHA-1 signature algorithms are not permitted for use and hence, cannot be registered with the HSM.

4.5. E. Erasing PHEFT2 HSM's Memory

Appendix E. Erasing PHEFT2 HSM's Memory

In order to securely erase (permanently) all user key material on the PHEFT2, for e.g. prior to disposal, the user should press the decommission button on the rear of the unit. An alternative route to decommission that can be performed remotely or if the red switch is damaged is to use the `hsm decommission` command.

> ! **Important Note**: All data will be lost once erase memory is perform. All keys, passwords, and configurations will be erased.

There are 2 ways to erase the PHEFT2 HSM's memory:

Emergency Decommission Button

Press and release the small red button, recessed behind the grille on the back panel. You need not power on the appliance nor have the power cable connected. You will need a small screw-driver or other tool to reach the Emergency Decommission button. This is intentional, to preclude accidental pressing of that button. You must re-initialize it before you can begin using it again. All the contents of Luna EFT are lost. You must power cycle the PHEFT2 appliance by pressing the momentary-contact START/STOP switch on the back panel of the system.

hsm decommission Command

1. Use the `hsm decommission` command to reset the appliance to the settings created at the factory. Please follow the following step:
2. From LunaSH command, login as admin. Please refer to C.1 Admin User Login LunaSH.
3. Then login as EFT admin. Please refer to C.6.1 EFT Admin Login Luna EFT Administration Console.
4. Perform the following command:
   `lunash:> hsm decommission`
5. Lunash will prompt warning to ask you confirmation.
6. Enter "proceed" to continue decommission.
7. The message "hsm decommission successful" is displayed upon successful completion of the operation.

4.6. F. Destructive Actions

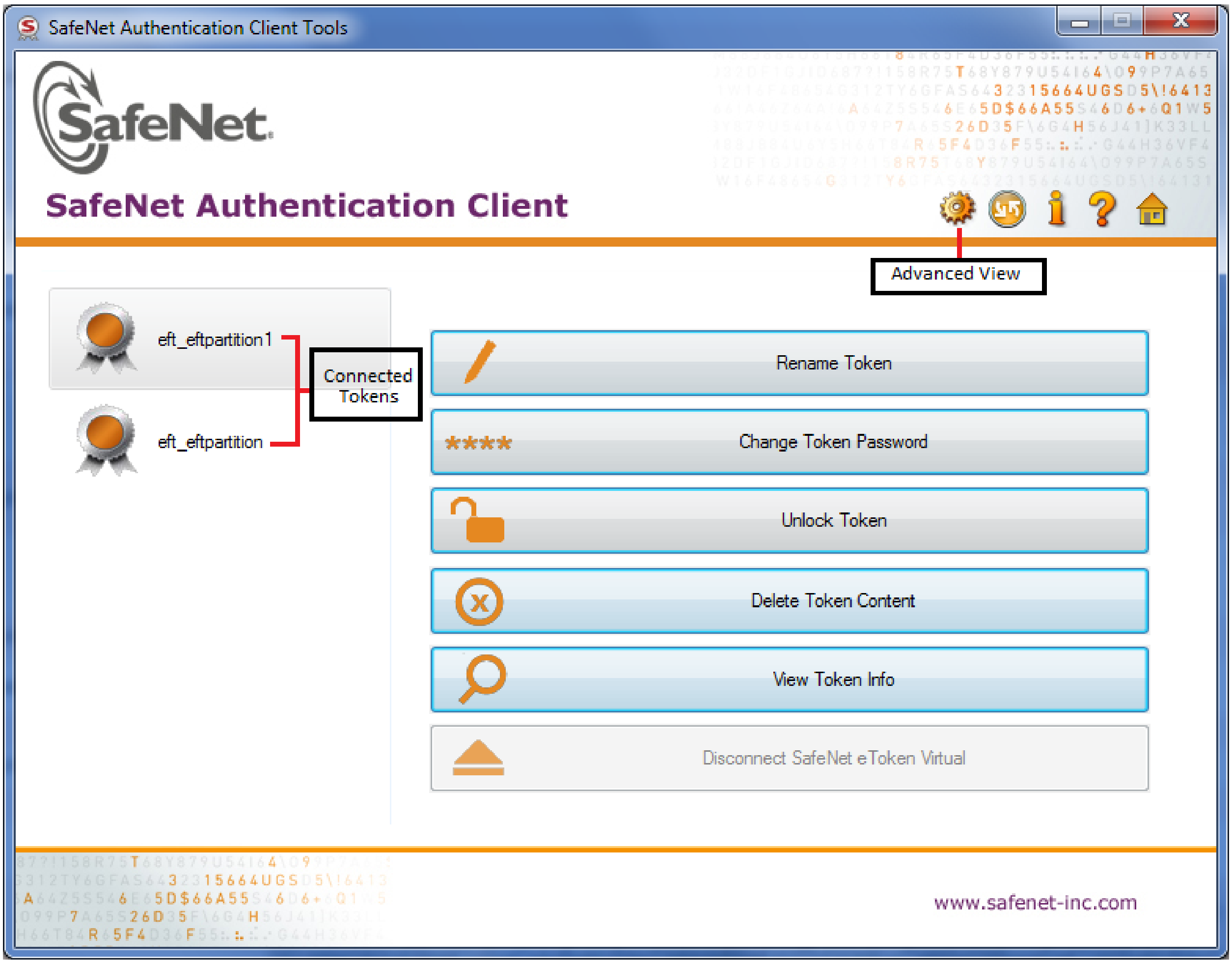
Various operations on the PHEFT2 appliance are intended to make HSM contents unavailable to potential intruders. The effect of those actions are summarized in the following table.

Event	Cryptographic Module	admin settings	EFT Admin settings	Partition Owner settings	EFT Auditor settings	Audit Logs	OS Users
PCI Mode Toggle	Cleared	Retained	Cleared	Cleared	Cleared	Retained	Retained
Decommission (hsm decommission command)	Cleared	Cleared	Cleared	Cleared	Cleared	Cleared	Cleared
Emergency Decommission Button	Cleared	Retained	Inaccessible	Cleared	Inaccessible	Retained	Retained
HSM initialization (hsm eftinit)	Cleared	Retained	Cleared	Cleared	Cleared	Retained	Retained
HSM Tamper	Cleared	Retained	Inaccessible	Cleared	Inaccessible	Retained	Retained
Software Upgrade (PL upgrade)	Cleared	Cleared	Cleared	Cleared	Cleared	Retained	Retained

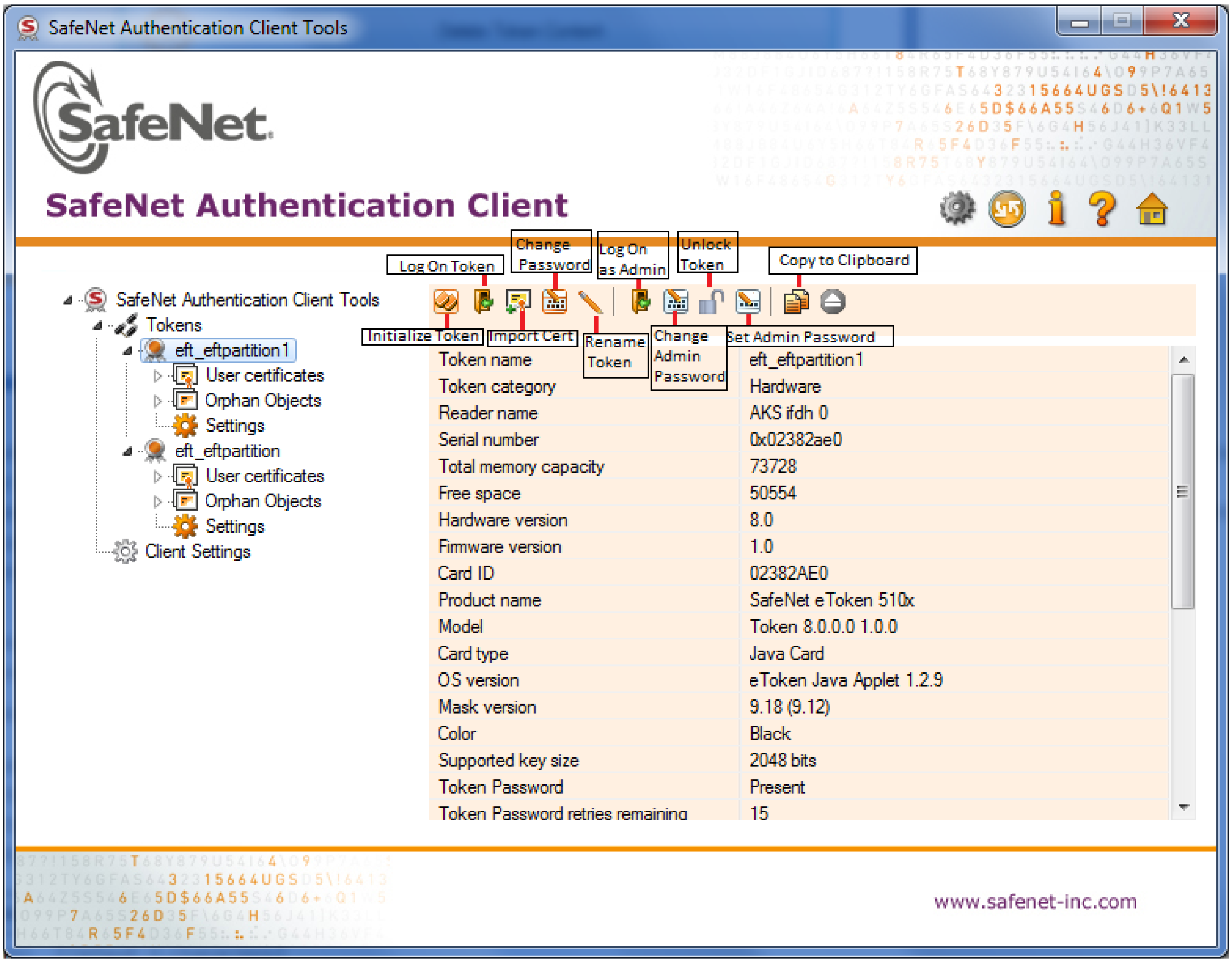
- **Cleared** - All the contents are erased.
- **Retained** - No impact. The settings remain as is.
- **Inaccessible** - No longer accessible.

4.7. G. SafeNet Authentication Client (SAC)

SafeNet Authentication Client (SAC) installer is come together with PHEFT2 HSM. SAC allows user to performs token management. The image below is the SAC main menu that enables you to control the use of your tokens:



The image below is the SafeNet Authentication Client Advanced View which provides additional token management functions:



Logging On to the Token as a User

You must log on to the token before you can use or change its token content.

To log on as a user:

1. Open the SafeNet Authentication Client Tools Advanced view.
2. In the left pane, select the node of the required token.
3. In the right pane, click the **Log On Token** icon.
4. The **Token Logon** window opens.
5. Enter the **Token Password**, and click **OK**.
6. You are logged on to the token.

Changing Token Password

SafeNet eTokens are supplied with an initial default token password. In most organizations, the initial token password is "1234567890". When a token password has been changed, the new password is used for all token applications involving the token. It is the user’s responsibility to remember the token password. Without it, the token cannot be used.

To change password for a token:

1. Open SafeNet Authentication Client Tools.
2. In the left pane, select the required token.
3. Select **Change Token Password** on the right pane.
4. Enter the **Current Token Password**. If this is your first time entering the token password, enter "1234567890".
5. Enter a **New Token Password**, then fill in the **Confirm Password** field.

Setting Token Password Quality

The password quality feature enables the administrator to set certain complexity and usage requirements for token passwords.

To set password quality for a token:

1. Open the SafeNet Authentication Client Tools Advanced view.
2. In the left pane, expand the node of the required token, and select **Settings**.
3. In the right pane, select the **Password Quality** tab.
4. Change the **Password Quality** settings, and click **Save**.

Setting Password Quality in Client Settings

Client settings are parameters that are saved to the computer and apply to all tokens that are initialized on the computer after the settings have been configured.

To set the password quality:

1. Open SafeNet Authentication Client Tools Advanced view.
2. In the left pane, select **Client Settings**.
3. In the right pane, select the **Password Quality** tab.

4. Change the **Password Quality** settings, and click **Save**.

Deleting Token Content

Objects on your token can include data objects (profiles), keys, and CA or user certificates. Your system configuration determines which objects are deletable. The **Delete Token Content** function deletes all deletable objects on your token. Non-deletable objects are not removed from the token. The function does not change settings on the token, such as password quality requirements.

To delete the token content:

1. Open SafeNet Authentication Client Tools.
2. In the left pane, select the required token.
3. In the right pane, select **Delete Token Content**.
4. The **Token Logon** window opens.
5. Enter the **Token Password**, and click **OK**.
6. The **Delete Token Content** window opens, prompting you to confirm the delete action.
7. To continue with the delete process, click **OK**.
8. The **Delete Token Content** window opens, confirming that the token content was deleted successfully.
9. Click **OK** to finish.

Logging On to the Token as an Administrator

If an administrator password was set on the token during token initialization, and the user forgets the token password, use the administrator password to unlock the token by setting a new token password. We recommend initializing all supported tokens with an administrator password.

To log on to a token as an administrator:

1. Open SafeNet Authentication Client Tools Advanced view.
2. In the left pane, select the node of the required token.
3. In the right pane, click the **Log On as Administrator** icon.
4. The **Administrator Logon** window opens.
5. Enter the token’s **Administrator Password**, and click **OK**.

6. You are logged on as an administrator.

Changing Administrator Password

If you are logged on to a token as an administrator, you can change the token's administrator password.

To change the administrator password:

1. Open SafeNet Authentication Client Tools Advanced view.
2. In the left pane, select the node of the required token.
3. In the right pane, click the **Change Administrator Password** icon.
4. The **Change Administrator Password** window opens.
5. Enter the current administrator password in the **Current Administrator Password** field.
6. Enter the new password in the **New Administrator Password** and **Confirm Password** fields.
7. Click **OK**. A message confirms that the password was changed successfully.
8. Click **OK**.

> **Note**: If an incorrect Administrator Password is entered more than a pre-defined number of times, the token becomes locked.

Unlocking a Token by an Administrator

If you are logged on to a token as an administrator, you can unlock the token by setting a new token password.

To unlock a token by setting a new token password:

1. Open SafeNet Authentication Client Tools Advanced view.
2. In the left pane, select the node of the required token.
3. In the right pane, click the **Set Token Password** icon.
4. The **Administrator Logon** window opens.
5. Enter the **Administrator Password**, and click **OK**.
6. The **Set Token Password** window opens.
7. Enter a new token password in the **New Password** and **Confirm Password** fields.
8. Set the **Logon retries before token is locked** field to the required number. Click **OK**.

9. A message confirms that the token password was changed successfully. Click **OK**.

10. The token is unlocked, and the user can now log on with the new token password.